

AGRI VISION 2025

1. INTRODUCTION :

Agriculture plays a vital role in the economy of Assam contributing 35% to the State Domestic Product in 1996-97 at constant (1980-81) price. The Government has, therefore, assigned very high priority to agriculture. This was reflected through a quantum jump in rice production to 39 lakh MT in 1999-2000 from the level of 31.55 lakh MT and 33.83 lakh MT in 1998-99 and 1997-98, respectively. This was achieved through creation of assured Shallow Tube Well irrigation with assistance of World Bank (ARIASP) and NABARD (SKY) scheme. Although regular flood, drought, heavy population pressure on land and infrastructural weakness are impediments to growth, yet the farmers have started to increase production through technological innovations and appropriate Government policies. During the next quarter century, the tempo of raising productivity and production must be accelerated along with integration of our economy with outside economies necessitated in the aftermath of liberalization.

In the premises of the above background a VISION PLAN through 2025 A.D. has been proposed keeping food security, employment generation and sustainability of agricultural production in its core. Thus a vision statement for Assam's Agriculture is crafted in the following words.

SUSTAINABLE AGRICULTURE FOR ENOUGH FOOD, EMPLOYMENT AND WEALTH

In order to achieve the above vision, areas of intervention that are to be addressed are discussed in this report. But before going to these, the profile of agricultural production base of the State and current status are presented in a brief manner.

AREAS OF INTERVENTION :

Sustainable agriculture is the key to keep up the level of production in the future. Issues related to this are discussed. Unemployment is another important area that is to be addressed in the agriculture sector. The industrial base of the State being narrow, agriculture sector must absorb the ever-growing labour force in the State. Self-sufficiency of food grains, oilseeds and horticultural products is the greatest necessity of the hour. However, excessive population increase has worsened the food situation. Although it is not discussed specifically in this report, serious efforts are necessary to mitigate the demographic problem. Introduction of large-scale shallow tube well irrigation in the State and future plans have reinforced the confidence of the State to wipe out deficit of rice from the next financial year. In this respect the strategy is to give stress on summer rice production with the irrigation potentiality created under shallow tube well programmes. Introduction of hybrid seeds under summer rice will augment and stabilize rice production. Emphasis on cultivation of scented rice and other varieties suitable for snacks during Sali season holds great promise for exports. A target of production of 10 million tonnes of rice has been fixed in 2007-08 to synchronize with the celebration of 60 years of India's independence. But the weakness of the State in this respect is the absence of a modern seed industry which is to be seriously thought of. Nutrient supply for targetted higher production of crops taking a judicious approach of Integrated Nutrient Management will be a challenging area of research in future. Use of green manure crops, bio-fertilizer and other organic manure to stabilize the agro-system shall be the focus of attention. The position of pulses production is far from satisfactory due to inherent lack of soil potentiality and absence of breakthrough in pulse research even in national level. However, strategies have been given to decrease deficit in pulse requirement. In respect of oil seeds, although the State is deficit, there is scope to mop up the deficit and become surplus. Necessary approaches in this regard are also given. In case of horticulture, the State's position is good. The main thrust here will be value addition inviting participation of the private sector. Floriculture is an entirely new field. The potentiality of floriculture can be tapped to augment export earning and mitigate unemployment for which modern infrastructures are to be built. In order to achieve self-sufficiency, agricultural mechanization is the crying need of the hour. Issues related to mechanization are also discussed. Requirement of inputs and other resources are also presented in this report. Other related issues in respect of infrastructure, training, WTO regime and role of Information Technology in Assam's vision in the next quarter century are also presented.

However, increasing population pressure on land will result in shrinkage of cultivated area. By 2025 AD estimated additional population will be about 178 lakh. This population will require about 2 lakh hectares for habitation alone. There will be huge requirement of food for this additional population. In order to meet this requirement, vertical increase of production will remain a pressing necessity during next 25 years.

2. AGRICULTURAL PRODUCTION BASE :

2.1 DISTRIBUTION AND OWNERSHIP CONCENTRATION OF LAND (ASSET) :

Information provided by the Directorate of Economics and Statistics on the number, area and average size of operational holding in Assam in 1990-1991 paint a grim picture of distribution of land among different size classes of operational holdings, (Economic Survey Assam, 1997-1998). Out of 25.2 lakh operational holdings, covering an area of 32 lakh hectares, marginal holding of one or less than one hectare of land accounted for 60.3 percent of total holdings which was only 19 percent of total operated area. Average size of holding was 1.27 hectare. It shows a skewed distribution pattern of land and asset among the farmers of Assam. The situation aggravates further with passage of time resulting in impoverishment of the farming community as consequence of fragmentation of holdings. However, no published data are available on pattern of concentration of land in holdings of different size classes. If control of assets leads to power and access to educational and income earning opportunities, it calls for speedy land reform to restore self confidence among the marginal farmers of the state.

However, large scale organization of Field Management Committees (FMCs) among farmers of Assam to protect them from economic vulnerability has contributed towards instilling self confidence and enabling them to participate in agricultural development process. There are about 25,000 FMCs in the state now.

2.2 TOTAL AREA AND CLASSIFICATION OF AREA OF ASSAM, 1995-1996 :

a.	Total geographical area	78,43,800 Hect.
b.	Forest	20,12,319 Hect.
c.	Area not available for cultivation	24,55,613 Hect.
d.	Net area sown	25,45,000 Hect.
e.	Total cropped area	36,94,000 Hect.
f.	Area sown more than once	12,15,195 Hect.
g.	Cultivable waste land	88,043 Hect.

Out of the above net area sown, 3 to 5 lakh hectares are affected by flood to various degrees annually.

3. PRESENT STATUS OF AGRICULTURE :

3.1 AREA AND PRODUCTION TREND OF RICE, WHEAT, PULSE AND OILSEEDS :

Table – I

Year	Rice			Wheat		
	Area	Production	Yield/ha.	Area	Production	Yield/ha.
1951-52	15.47	13.98	919	0.018	0.016	896
1960-61	17.16	16.33	968	0.036	0.030	824
1970-71	19.74	19.86	1022	0.200	0.110	583
1980-81	22.28	24.59	1120	0.910	1.060	11.58
1990-91	25.26	32.70	1313	0.830	1.040	12.48
1995-96	25.39	33.90	1354	0.860	0.950	11.07
1996-97	25.28	33.28	1336	0.880	1.170	13.32
1997-98	25.26	33.83	1359	0.860	1.000	12.99
1998-99	24.54	32.55	1326	0.900	0.910	10.11
1999-2000	26.52	39.00	1470	0.850	1.200	11.75

Table-II

Year	Pulse			Oilseeds		
	Area	Production	Yield/ha.	Area	Production	Yield/ha.
1951-52	0.81	0.28	347	1.25	0.53	434
1960-61	0.74	0.26	356	1.25	0.48	386
1970-71	0.84	0.32	379	1.49	0.62	416
1980-81	1.11	0.46	432	2.17	1.05	485
1990-91	1.21	0.52	433	3.40	1.90	559

1995-96	1.18	0.63	534	3.33	1.87	560
1996-97	1.30	0.75	578	3.35	1.84	551
1997-98	1.28	0.71	552	3.46	2.04	589
1998-99	1.38	0.74	536	3.44	1.75	508
1999-2000	1.50	1.00	666	3.84	2.41	627

Area = Lakh hectares; Production = Lakh MT; Yield/ha. = Kg.

3.2 HORTICULTURE :

The total area under major horticultural crops in the state is 5,42,942 hectares out of which fruit crops occupy an area of 1,04,816 hect. , spices 78,633 hect., tuber 86,391 hect., nut crops 94,623 hect. and vegetables occupy 1,78,479 hectares.

The existing area and production of major horticultural crops in Assam and the area (district) where the crops are predominantly grown are indicated below :

Table III

Sl. No.	Name of the crop	Area in hect.	Production in M.T.	Average yield kg./ha.	District where predominantly grown
1.	Banana	41,885	5,81,884	13,892	Barpeta, Kamrup, Nagaon, Cachar, Nalbari, Goalpara, Jorhat, Golaghat, Sonitpur, Sibsagar, Tinsukia
2.	Pineapple	13,582	2,08,993	15,387	Kamrup, Nagaon, Karbi Anglong, N.C. Hills, Cachar
3.	Orange	5,720	67,659	11,820	Tinsukia, Karbi Anglong, North Cachar, Kamrup, Goalpara, Darrang, Dibrugarh
4.	Papaya	7,347	1,08,479	14,765	Nagaon, Darrang, Kamrup, Karbi Anglong, N. Cachar, Nalbari, Barpeta, Sonitpur, Dhubri
5.	Assam lemon	8,705	47,233	5,426	All districts of Assam
6.	Guava	3,681	47,611	12,934	All districts of Assam
7.	Litchi	4,028	16,815	4,175	All districts of Assam-mainly Kamrup, Sonitpur and Bongaigaon
8.	Jack-fruit	17,298	1,53,645	8,882	All districts of Assam
9.	Mango	2,570	17,173	6,682	All districts of Assam
Total :		1,04,816	12,49,492	1,192	

SPICES :

1.	Chilli	14,724	9,619	658	Dhubri, Barpeta, Darrang, Nagaon
2.	Turmeric	10,729	7,416	691	Kamrup, Darrang, Nagaon, Barpeta, Sonitpur, Nalbari, Bongaigaon
3.	Onion	8,083	18,341	2261	Barpeta, Nagaon, Dhubri
4.	Ginger	16,244	1,13,771	7280	N.C. Hills, Cachar, Karbi Anglong, Kamrup, Barpeta, Sonitpur, Nagaon

5.	Coriander	19,346	14,354	742	Dhubri, Nalbari, Kamrup, Barpeta, Sonitpur, Nagaon, Morigaon, Darrang, Goalpara
6.	Garlic	7,430	21,694	2920	Dhubri, Kamrup, Barpeta, Nagaon, Lakhimpur, Goalpara, Darrang
7.	Black pepper	2,077	2,810	1353	Jorhat, Sibsagar, Kamrup, Nagaon, Golaghat, Dibrugarh, Cachar, Barpeta
Total :		78,633	1,88,076	2392	

NUT CROPS :

1.	Coconut	20,166	1,49,866 (in '000 nos)	69 nos. (per tree)	Nagaon, Nalbari, Kamrup, Morigaon
2.	Areca nut	74,457	55,355 (dry nuts)	131 nos. (per tree)	All districts of Assam
Total :		94,623			

TUBER CROPS :

1.	Potato	76,958	6,11,077	9740	Barpeta, Darrang, Kamrup, Sonitpur, Nagaon, Dhubri, Nalbari
2.	Sweet potato	9,43	32,437	3438	Dhubri, Kamrup,
3.	Tapioca	2,798	13,261	4739`	Kokrajhar, Nalbari, Darrang, Karbi Anglong, Goalpara
Total :		86,391	6,56,775	7602	

VEGETABLES :

1.	Kharif	56,857	5,37,786	9,411	All districts of Assam
2.	Rabi Veg.	1,21,622	18,45,435	15,230	All districts of Assam
Total :		1,78,479	23,83,221	13,205	

3.3 IRRIGATION :

From the time of independence till 1995-96, 61,428 Shallow Tube Wells were installed in Assam. During the next four years, Assam saw a quantum jump both in terms of number of tube wells installed and area irrigated. Financial assistance was availed of from the World Bank (ARIASP) and NABARD (SKY) resulting in installation of an additional 62,250 Shallow Tube Wells within March 2000 (32,250 under ARIASP and 30,000 under SKY). Another 70,000 Shallow Tube Wells are being installed under NABARD and expected to be completed soon and 15,000 under World Bank by March 2001. It means that by March 2001 installation of a total number of 1,47,250 Shallow Tube Wells will be completed which will provide assured irrigation to 2,94,500 ha. land and including the Shallow Tube Wells installed prior to 1995-96, the total assured irrigation will be available in 5.27 lakh ha.

3.4 TREND OF FERTILIZER CONSUMPTION :

The trend of fertilizer consumption in Assam in terms of nutrients (N, P₂O₅, K₂O) per hectare is shown below:

Year

Kg./Ha.

-----	-----
1971-72	2.4
1981-82	3.3
1991-92	10.4
1995-96	13.0
1996-97	14.2
1997-98	18.2
1998-99	21.1
1999-2000	29.3

The projected consumption of fertilizer nutrients per hectare has been fixed at 50 kg in 2000-2001 and 70 kg from 2005-2006 onwards. It is not the intention of the Agriculture Department to go beyond 70 kg./ha. and additional requirement would be met through Integrated Nutrient Management like use of organic manures, bio-fertilizers and green manuring to maintain soil health.

3.4 SEEDS :

Assam was once wholly dependent on outside supply for meeting its own requirement of seed till Assam State Seed Certification Agency (ASSCA) was constituted in the year 1985. But till now the State could not achieve self-sufficiency in production of its own requirement of seeds except paddy and mustard due to infrastructural inadequacy. It is essential that the seeds required for production programme of the State is produced within the State to ensure timely supply of seeds in quality and quantity to the farmers. From the experience it can be inferred that the Assam Seeds Corporation Ltd. may not be able to produce all the required quantities of certified seeds of various crop, whose seeds can be produced in the State, due to their infrastructural inadequacy including the finance. About 40 Nos. of Seed farms belonging to the Department of Agriculture have been proposed to be leased out to private entrepreneurs with an idea of producing certified seeds in accordance with the requirement of the State. The seed farms which are already leased out have started producing certified seeds of the varieties which are locally in demand. Thus seed production and marketing through seed merchants / traders/ entrepreneurs are to be encouraged for attaining self-sufficiency in seed production in the State.

3.5 AGRICULTURAL MECHANISATION :

The growth of mechanization in Assam is slow. However, after introduction of large scale Shallow Tube Well irrigation, requirement of farm power has gone up. The present position of mechanization is as follows:

The following are the estimates of power for raising agriculture crops in Assam..

Category of power	Amount (HP per hectare)
-----	-----
Human power	0.09
Animal power	0.204
Mechanical power	0.008
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Total available power	0.302

The current power availability to the farmers of the State is barely 0.3 HP per hect. which has become one of the constraints towards increase of area under double or multiple cropping.

It is obvious that farm mechanization is imperative to supplement the poor bullock draught power which is grossly inadequate to meet the challenging task of covering additional areas under cultivation.

The present status of machineries in the state reflects a picture of grossly inadequate availability of mechanical power.

a.	Total Nos. of M. Wheel Tractors (Four Wheel Tractor) including Govt. sector.	678
b.	Total Nos. of Power Tillers (Two wheel Tractors) including Govt. sector.	5,002

c.	Total Nos. of M.B.plough	1,10,477
d.	Total Nos. of paddy weeder	17,926
e.	Total Nos. of Dry land weeder	2,032
f.	Seed Drill	722
g.	Total Nos. of Bullock Drawn puddler	149
h.	Total Nos. of paddy cultivator	4,723
i.	Total Nos. of Harrow	465
j.	Total Nos. of other implements	1,67,449

The density of Tractor is 0.26 per thousand hectare and that of Power Tiller is 0.51 per thousand hectare.

4. FUTURE OUTLOOK :

4.1 SUSTAINABLE AGRICULTURE :

Agriculture depends on natural environment of a place like land, air, flora, fauna and water. But destruction of natural resources and environment introduces of limiting factor for agricultural production. It is important for us to harvest crops without decrease in yield over time so as to feed the growing population. Therefore, in the present context of rapid degradation of natural environment, the question of sustainability refers to “ meeting the needs of the present generation without compromising the needs of future generations”.

Degradation of environment has threatened major age old eco-systems of Assam like swamps and drainage systems. Water bodies are progressively filled up or choked. Many land and aquatic life forms are on the brink of extinction, which affect crop production. The forest cover in the state is fast dwindling which has caused man-made climatic aberrations (as experienced now) and degradation of soil. Clearing of forest encourages soil erosion and run-off without affording any time for percolation of rain water to deeper soil layers for recharge of ground water. Thus ground water hydrology is affected.

The destruction of forest cover in adjoining hills of Assam resulted in silting up of water bodies and water courses. This besides inundating a vast area of valley and causing flash flood in foot hills, is also rendering thousands of hectares of land unproductive due to deposition of sand. The inundation and flash flood is causing huge loss of crop production together with destruction and damage of infrastructure and human habitation.

Occurrence of drought is also a regular phenomenon due to degradation of natural eco-system and irregular breaks in rainfall. The extent of loss of crop production due to drought cannot be underestimated and leaves an impact on the economy of the farming community.

High population pressure on land, besides damaging the environment, has changed land use pattern from agricultural to non- agricultural use resulting in shrinkage of cultivable land. During the next 25 years projected population increase is about 178 lakh, which is more than half of the present population of Assam. The gradual settlement of this additional population will take away about 2 lakh hectares from the existing cultivated area for habitation requirements in the next quarter century.

In the light of the above, the problem of sustainable agriculture can be addressed with the combined strategies of conservation of natural resources and poverty alleviation, which are outlined below.

CONSERVATION OF NATURAL RESOURCES STRATEGIES :

1. Protection of forest and soil.
2. Conservation of rainwater.
3. Development of natural eco-systems like swamps and beels.

POVERTY ALLEVIATION STRATEGY :

1. Sustainable methods of agricultural production.
2. Women's involvement.

3. Land reform and access to inputs for marginal and small farmers.

STRATEGIES FOR CONSERVATION OF NATURAL RESOURCES :

1. PROTECTION OF FOREST AND SOIL :

The existing forest area is to be protected at any cost for economic survival of the people of Assam. Along with the forests, the soil wealth is also to be protected adopting soil conservation measures. However, both types of protection is to be converted to a people's conservation movement. Scarcity of firewood among the poor leads to destruction of forests. Introduction of farm forestry into agriculture would ease the demand of fuel and fodder for animals and thereby help preserve our forest.

2. CONSERVATION OF RAIN WATER :

In-situ conservation of rainwater is most important. The National Watershed Development Project for Rainfed Areas (NWDPPRA) presently implemented by Agriculture Department as centrally sponsored scheme is doing useful work in Assam. However, greater stress may be given for construction of rainwater harvesting structures for recharging ground water besides taking advantage of part time fish culture and irrigation from them.

3. DEVELOPMENT OF NATURAL ECOSYSTEMS LIKE SWAMPS AND BEELS :

Progressive choking up of swamps and beels must be stopped. A special programme may be started to decongest the water bodies and to remove silt from the swamps or beels taking assistance if necessary from international funding agencies. The DRDAs also may play a role in this respect. This will help improve environment of our natural heritage, create recreation sites and recharge ground water besides providing shelter to fish and birds.

STRATEGIES FOR POVERTY ALLEVIATION :

1. SUSTAINABLE METHODS OF AGRICULTURAL PRODUCTION :

Excessive use of chemicals in agriculture tends to pollute land and water. This may be avoided by using Integrated Pest Management (IPM), Integrated Nutrient Management (INM) and organic farming. The Assam Agriculture Department has already introduced IPM technique in the State's agriculture and imparted training to field officers. This is to be vigorously pursued now. Infrastructural facilities for supply of bio-control agent is to be widened. Ground water used extensively now for irrigation in Assam is to be monitored for content of any toxic substances so that they cannot pollute land and crops. Recent discovery of fluoride in excessive concentration in the ground water of Karbi Anglong district is a case in point, which is a health hazard.

2. WOMEN'S INVOLVEMENT :

Women play a pivotal role in management of resources of household and community. Fuel and water supplies, backyard poultry, weaving, kitchen gardening and a lot of farm activities from planting to harvesting, drying, milling etc. are taken care of by women besides attending to their regular work of cooking, cleaning and looking after the family members. Generally women are left out of training programmes designed for skill development required after introduction of new technology or innovations. Child nutrition, hygiene and lowering fertility rates are related to women's education. But lower educational attainment of women pose a problem in the fight against poverty perpetuating process. Women's skill upgradation and education will remain a precondition for successful environment management in Assam for a long time to come. Women are to be always at the center of sustainable agricultural programmes.

3. LAND REFORM AND ACCESS TO INPUTS FOR MARGINAL AND SMALL FARMERS :

Environmental degradation is related to poverty. Raising the productivity of existing resources of the poor is a very important criterion for preservation of environment and sustainable agriculture. Land reform to transfer rights to sharecroppers and making availability of agricultural inputs including credit easy to marginal and small farmers would go a long way in raising agricultural productivity and income of the poor.

4.2 SELF SUFFICIENCY IN FOOD :

A. FOOD GRAIN PRODUCTION :

Food grain crop consists of Rice, Wheat & Pulses. Requirement of food is dependent on population. Taking a growth rate of population as 2.11 P.C. from 2000-01 to 2009-2010 and 2.00 P.C. from 2010 – 2011 to 2025 – 2026 the projected population of the state as well as requirements of food grains will be as under :

Table IV

Year	Population	Adult equivalent	Rice (in MT)	Wheat (in MT)	Pulses (in MT)	Total (in MT)
2000-2001	27159300	21727440	4282478	570997	570997	5424472
2001-2002	27732361	22185889	4372839	583045	583045	5538929
2002-2003	28317514	22654011	4465106	595347	595347	5655800
2003-2004	28915014	23132011	4559319	607909	607909	5775137
2004-2005	29525120	23620096	4655521	620736	620736	5896993
2005-2006	30148100	24118480	4753752	633834	633834	6021420
2006-2007	30784225	24627380	4854057	647208	647208	6148473
2007-2008	31433772	25147018	4956477	660864	660864	6148473
2008-2009	32097025	25677620	5061059	674808	674808	6410675
2009-2010	32774272	26219418	5167847	689046	689046	6545939
2010-2011	33429758	26743806	5271204	702827	702827	6676858
2011-2012	34098353	27278682	5376628	716884	716884	6810396
2012-2013	34780320	27824256	5484161	731221	731221	6946603
2013-2014	35475926	28380741	5593844	745846	745846	7085536
2014-2015	36185445	28948356	5705721	760763	760763	7227247
2015-2016	36909154	29527323	5819835	775978	775978	7371791
2016-2017	37647337	30117870	5936232	791498	791498	7519228
2017-2018	38400284	30720227	6054957	807328	807328	7669613
2018-2019	39168289	31334631	6176056	823474	823474	7823004
2019-2020	39951655	31961324	6299577	839944	839944	7979465
2020-2021	40750688	32600551	6425569	856742	856742	8139053
2021-2022	41565702	33252562	6554080	873877	873877	8301834
2022-2023	42397016	33917613	6685161	891355	891355	8467871
2023-2024	43244956	34595965	6818865	909182	909182	8637229
2024-2025	44109855	35287884	6955242	927366	927366	8809974
2025-2026	44992053	35993642	7094347	945913	945913	8986173

The state of Assam comprising 21 number of plain districts and two number of hill districts fall under high potential zone for raising agricultural productivity. Self-sufficiency in production of rice and to reduce to a possible extent the shortfall in

other items of food grains viz., wheat and pulse are the primary objectives of the state. Having considered the rabi season as the vital period for the growth of agriculture in the state due to uncertainties prevailing during kharif season, the Government have set priorities in the development of infrastructure in the area of minor irrigation, agricultural mechanization, enhancement of fertilizer consumption rate and production of certified seeds.

Among the food grain crops, rice is the principal crop grown in three seasons viz., Autumn (Ahu), Winter (Sali) and Summer (Boro & Early Ahu). Production of food grains in the state is primarily dependent on production of Winter rice which covers about 72 P.C. of the rice growing area. The crop is sown in the rainy season and is, therefore, subject to heavy flood almost every year. Because of the unstable nature of production, the area under summer rice is on the rise. With the higher productivity of summer rice among all rices grown in the state, Assam recorded an impressive increase in rice production during 1999-2000 producing about 39 lakh MT. If 20 P.C. of production is deducted for storage loss, animal feed etc., Assam had marginal deficit in rice during 1999-2000. But with the launching of the STW programme under World Bank (ARIASP) and NABARD (SKY) assistance, there was increase of about one lakh hectare area under summer rice which increased production. The scope of the STW programme is currently being expanded further and it can be expected that both productivity and production of total rice will increase sharply in Assam.

The productivity of rice in Assam is around 1500 kg/ha against national average of approximately 2000 kg/ha during 1999-2000. Required productivity over the next 25 years to maintain self sufficiency in rice are projected below :

Year	* Gross rice area in lakh hect.	Productivity kg./hect.	Total production in lakh MT
2000-01	27	2200	59.40
2005-06	30	2500	75.00
2006-07	30	3340	100.20
2010-11	30	3500	105.00
2015-16	30	3800	114.00
2020-21	30	4000	120.00
2025-26	30	4500	135.00

* Area likely to increase with the expansion of irrigation.

STRATEGIES TO INCREASE PRODUCTIVITY OF RICE :

1. Expansion of assured irrigation to cover 15 lakh hect. at the end of 2010.
2. Increase the fertilizer consumption to 70 kg./hect. by 2010.
3. Increase in seed replacement rate to 12% including Hybrid Rice.
4. Introduction of mechanization.
5. Adoption of intensive IPM and INM in rice.

In consideration of above facts, it can be safely said that Assam will attain self-sufficiency in rice from 2000-2001 onwards.

B. PULSE :

Pulses are rich sources of protein in our diet. They can be cultivated in various cropping systems without disturbing the main crops in the rotation. Besides, they can fix atmospheric nitrogen in the soil and hence have become very important in the context of developing sustainable agriculture in Assam. Besides intensification of agricultural activities with ground water utilization at present times also warrants pulse cultivation to stabilize soil health.

Production of pulses is around 80,000 MT at present which can meet a little over 20% requirement of the State. The common kharif pulses are Arahar, Black gram and Green gram and rabi pulses are Lentil, Pea, Black gram, Green gram, Chickpea and Rajmah. Low production of pulses in the State is attributed to (a). Soil acidity, (b) lack of suitable improved seed and (c) Lack of infrastructure for storage.

Requirement of pulses shown at 5 year intervals from 2000-2001 to 2020-2021 :

Year -----	Requirement calculated @ 60 gm/adult/day in lakh MT -----	Requirement calculated @ 30 gm/adult/day + 20% for seed, feed and storage loss -----
2000-01	4.76	2.85
2005-06	5.28	3.17
2010-11	5.87	3.53
2015-16	6.51	3.90
2020-21	7.23	4.34
2025-26	7.88	4.73

Estimate of reduced requirement of pulses has been shown in the above table in consideration of the following :

1. Very high prices of pulses force the people to reduce consumption.
2. Fish being a common ingredient of our diet, supplement protein requirement substantially.
3. People in many rural areas of Assam usually meet their pulse requirement partially through dried country beans.

PULSE PRODUCTION PROGRAMME IN ASSAM :

The gap between requirement and production is met by importing pulses worth about Rs. 400 crores annually to the State. Therefore, removal of deficit has assumed great importance and endeavours would be made to decrease the gap between supply and demand to the extent possible. Cultivation of pulses is to be supported by the post harvest management like Dal Processing Plants and storage bins where rural youths can find employment over and above engagements in crop production.

STRATEGIES FOR PULSE CULTIVATION IN ASSAM :

1. Increase of area under rabi pulse by another 1.75 lakh hectares by 2025-26.
2. Increase of productivity from the present level of 600 kg/ha to 1200 kg/ha by 2025-26.
3. Increase of area under Arahar (Pigeon Pea) during kharif.
4. Popularization of modern pulse production technology like improved seeds and use of bio-fertilizers.
5. Introduction of rabi pulse as intermediate crop under irrigated ecology –

Cropping sequence to be followed :

Kharif Rice (June – Oct/Nov) ----- Rabi Pulse (Nov – Jan/Feb)
 Summer Rice (Jan/Feb – May/June)
 This can maintain soil health / fertility.

6. Production of pulse seeds through seed village concept.

Proposed targets for pulse production at 5 yearly intervals from 2000-01 to 2025-26 :

Year -----	Area target In lakh hect. -----	Production In lakh MT -----	Productivity In kg./ha. -----	Percentage of deficit -----
2000-01	1.50	0.90	600	68%
2005-06	2.00	1.40	700	56%
2010-11	2.50	1.88	750	47%
2015-16	2.75	2.34	850	40%
2020-21	3.00	3.00	1000	31%
2025-26	3.25	3.90	1200	18% (marginal)

C. OILSEED :

Rape and Mustard are major oil seed crops of Assam.. Other oil seed crops cultivated during the Rabi season are Sunflower, Groundnut and Nizer and Sesamum and soybean are cultivated during the kharif season. Rape and mustard cover about 2.25 lakh hectare in Assam. Coverage under other oilseed crops is not significant. Overall productivity of oil seeds in Assam is around 600 Kg per hectare which is almost half the national average. In Assam oil seeds are raised under rainfed condition with low level of modern agro technique. Therefore, productivity is low as compared to the national average. Thrust given on research by the A.A.U. has produced two varieties of Toria viz., TS-36 and TS-38 with yield potential of 1.5 MT per hectare. This is being pursued further. In the event of increase in the net irrigated area to a large extent, it is likely that rabi oil seeds will be cultivated more and more with irrigation. This will push up the productivity of oil seeds.

At present Assam produces around 60 percent of her total requirement. The vision of the state in oil seed production during the next 25 years is progressively narrowing down the deficit and then achieving surplus through a combination of modern production technology and improved service facilities supported by appropriate government policy. Higher oil seeds production will create more employment in the entire chain from production to processing and marketing.

Requirement of oil seeds of Assam at 5 yearly interval is given below :

Year -----	Requirement of oil seeds -----	Remarks (in lakh MT) -----
2000-01	3.81	<u>Requirement of oil include :</u>
2005-06	4.23	(1) 40 gms per adult per day.
2010-11	4.68	(2) Seed and wastage – Add. 20%.
2015-16	5.17	
2020-21	5.71	
2025-26	6.31	

Projected area, production, productivity and deficit/ surplus situation of oil seeds in Assam during the next 25 years is shown below :

Year -----	Area lakh hect. -----	Productivity Kg/ Hect -----	Production lakh MT -----	Percent deficit/ surplus -----
2000-01	4.00	650	2.60	(-) 32
2005-06	4.15	700	2.90	(-) 31
2010-11	4.25	800	3.40	(-) 28
2015-16	4.50	1000	4.50	(-) 13
2020-21	4.75	1200	5.70	0
2025-26	5.00	1500	7.50	(+) 19

Strategies and approaches of increasing oil seeds production :

1. Increase in area.
 - a. Expansion of area in monocropped Sali land.
 - b. Expansion of areas of non-traditional oil seeds like ground nut and sunflower.
2. Increase in productivity.

- a. Replacement of old varieties with the newly developed high yielding varieties.
 - b. Increase in fertilizer consumption.
 - c. Conversion of rainfed to irrigated oil seeds
3. Improvement in post harvest and marketing management.

D. HORTICULTURE :

The State of Assam has almost attained self-sufficiency in its requirement of cereals – more particularly rice and as the position stands today, the State is looking for export. Now all eyes are set on the protective food sector i.e. fruits, vegetables and other horticultural crops. On the basis of per capita requirement the State is marginally surplus both in fruits and vegetables. During 1998-99, the State had a population of 2.62 crores and on that basis the requirement is calculated at 6,88,657 MT in case of fruits and that of vegetables is 21,42,488 MT. Against this requirement, our present production of fruits and vegetables are 12,49,492 MT and 30,26,735 MT, respectively. The fate of this surplus quantity either becomes a part of heavy post harvest loss or becomes a component of the regular outflow to the neighbouring deficit states. Though the scenario is somewhat better, it is necessary to look for immediate strategy if we are to go ahead on the basis of increasing population, market demand, projected requirements in the coming years for sustainability, income and employment generation. The requirement of fruits, vegetables and spices for next 25 years i.e. from 2000 to 2025 is projected in Table - V below :

Table V
PROJECTED POPULATION AND
REQUIREMENT OF FRUITS, VEGETABLES & SPICES FROM 2000-2001 TO 2020-2021

Year	Projected population	Adult equivalent	Requirement in MT			Remarks
			Fruits	Vegetables	Spices	
2000-2001	2,71,59,300	2,17,27,440	7,13,746	20,20,544	79,305	
2005-2006	3,01,53,268	2,41,22,614	7,92,428	24,65,331	88,048	
2010-2011	3,34,77,288	2,67,81,826	8,79,783	27,37,103	97,754	
2015-2016	3,71,67,727	2,97,34,182	9,76,767	30,38,833	1,08,530	
2020-2021	4,12,69,997	3,30,11,997	10,84,444	33,73,826	1,20,494	
2025-1026	4,49,92,053	3,59,93,642	11,82,391	36,78,550	1,31,377	

From the above table it is clear that the requirement of fruits in the year 2025-2026 will be 11,82,391 MT. However, the above figure is the consumption requirement of adult equivalent of total population of the State. The production of fruits during 1998-99 was 12,49,492 MT against the consumption requirement of 6,88,657 MT indicating that approximately 45% of the production is either marketed / exported outside State or get wasted. There is ample opportunity to export fruits like Banana, Pineapple, Orange, Jackfruit, Litchi etc. as raw fruit, processed fruit and after primary processing to other States of India as well as outside the country. In fact, export of some commodities has already started across the border and this trade is likely to be expanded in near future. Considering all these above factors, the vision of 2025-2026 is outlined in Table VI. Increase in production is aimed by expansion of area presently remaining as cultivable waste in the upland and hilly tracts and by vertical increase of production.

So far as vegetables are concerned the requirement by 2025-2026 will be 36,78,550 MT against the current total production of 30,26,735 MT indicating that the State will be short by 6,51,815 MT if the production remains static over the years. However, as the vegetables are highly perishable commodities the wastage is more in comparison to other crops. Besides, the State of Assam is being surrounded by deficit states and countries, hence outflow is obvious. Taking into consideration of all the above factors the future production is targeted at 50% above the actual consumption requirement assuming that 20% will be lost on account of field wastage and transportation and 30% will be marketed outside that State – either as raw or as processed vegetables. The targeted area and production is given in Table VI. The increase in area under vegetables will be effected by bringing feasible mono-cropped area of Sali under vegetable cultivation during rabi season and

by increasing cropping intensity by introduction of kharif vegetables in typical and feasible flood free rabi vegetables areas as well as by increasing inter-cropping of kharif vegetables in areas covered by fruits and nuts.

There are wide varieties of medicinal and aromatic plants in the State and the State being a hot spot of bio-diversity, there is immense potentiality for production of different kinds of medicinal and aromatic plants. The Small Farmers' Agri-business Consortium (SFAC) will take a lead by contacting the Pharmaceutical and Aromatic Industries for contract production in first phase followed by pilot scale production of selected medicinal and aromatic plants and then the mass production of medicinal and aromatic plants accompanied with setting-up distillation plants and processing plants is targeted. Projected figures on area and production are shown in Table VI.

It is estimated that the projected requirement of spices during 2025 will be 1,31,377 MT (Table V) whereas our existing production is 1,88,076 MT. The present production is exceeding by 145% and a sizeable quantities of Chilly, Ginger & Coriander are going out of the State. Considering the present trend of growth, an achievable target will be 5,23,229 MT during 2025 i.e. an excess of 3,35,153 MT over the present production. To achieve this target, the strategy will be area expansion mostly as mixed crop with other plantations as well as by increasing production per unit area with improved inputs & technology. This sector will be treated as a major sector for employment generation. The projected area and production of Spices is shown in Table VI.

There is no authentic data on existing area and production of wide range of floriculture crops but according to rough estimate, this sector covers an area of 140 Hectare. Floriculture though has enough potentiality, is still at its infancy. Orchids, Anthurium, Gladiolus, Marigold, Rose have taken some roots on the commercial front but owing to rapid change in market trend more species like Jasmine, Lily, Gerbera may enter the market. Oil extraction from Jasmine, Tuberose, Rose may also be profitable venture. Thus considering the global trend and normally available rich stock of orchids, this crop will be put under commercial venture on priority followed by Anthurium, Gladioli, Roses, Jasmine and tuberose. The projected expansion of area and production till 2025-2026 has been assessed in Table VI.

The overall scenario of horticulture in Assam in near future by and large is expected to be highly commercial and market oriented – with the incorporation of horticulture based industrial growth – particularly in the form of establishment of cold storages, processing units like Ginger dehydration, Turmeric, Pepper processing, oleoresin / oil extraction units, semi-processed product manufacture, production of concentrates, distillation units for medicinal and aromatic plants, establishment of tissue culture units etc. with tremendous – employment generation. Finished and semi processed products may be marketed with appropriate arrangement with private sector with brand image having a wide marketing network in India. Generation of employment is also expected under area expansion thrust for crops like cashew nut, vegetables etc. establishment and setting up of primary processing unit, distillation unit, cold storage etc. The rapid stride in farm mechanization and ever expanding lucrative markets will invite entrepreneur to venture into these sectors. Establishment of sophisticated nurseries, Orchid growers network, TPS & other vegetable seeds production and marketing etc. are envisaged and likely to attract NGOs and entrepreneur to step into such sectors ensuring more and more employment. It is estimated that in case of cereals the employment generation is 150 to 200 man days per hectare where as horticultural sector can accommodate 350 to 2500 man days per hectare. In accordance with table II, we can expect generation of employment to the tune of 62 crore man days by 2025 even at a very modest estimate assuming that only 50% of the total targeted area will be explored by commercial units.

Table VI
TARGETTED AREA AND PRODUCTION OF FRUITS, VEGETABLES, SPICES, FLOWERS AND MEDICINAL & AROMATIC PLANTS
FOR NEXT 25 YEARS

Year	Fruit Crops		Vegetable Crops		Spices		Flowers		Medicinal & Aromatic Plants		Remarks
	Area (in Ha.)	Production (in MT)	Area (in Ha.)	Production (in MT)	Area (in Ha.)	Production (in MT)	Area (in Ha.)	Production (in MT)	Area (in Ha.)	Production (in MT)	
1998-99	1,04,816	12,49,492	2,64,870	30,26,735 (13.20MT/ha)	78,633	1,88,076	140	-	-	-	Increase in area in case of fruit is taken as 25% per annum, in case of vegetables the increase is aimed at 1% per annum and in case of spices the increase is projected as 1.5% per annum.
2000-01	1,10,057	12,20,684 (12.00MT/ha)	2,70,167	35,93,221 (13.30MT/ha)	80,991	1,94,378 (2.40MT/ha)	300	900	600	1,800	
2005-06	1,23,814	16,09,582 (13.00MT/ha)	2,83,675	39,71,450 (14.00MT/ha)	87,065	2,61,195 (3.00MT/ha)	800	4,000	800	2,400	
2010-11	1,39,291	19,50,074 (14.00MT/ha)	2,97,859	44,67,885 (15.00MT/ha)	93,595	3,27,583 (3.50MT/ha)	1,050	6,300	1,000	4,000	
2015-16	1,56,702	23,50,530 (15.00MT/ha)	3,12,751	50,04,016 (16.00MT/ha)	1,00,615	4,02,460 (4.00MT/ha)	1,200	8,400	1,200	4,800	
2020-21	1,76,289	28,20,620 (16.00MT/ha)	3,28,388	55,82,596 (17.00MT/ha)	1,08,161	4,32,644 (4.00MT/ha)	1,450	10,875	1,600	6,400	
2025-26	1,98,325	32,72,362 16.50MT/ha	3,44,807	58,61,719 (16.50MT/ha)	1,16,273	5,23,229 4.50MT/ha	1,750	14,000	2,100	9,450	

4.3 HILL AGRICULTURE :

Karbi Anglong and North Cachar Hills are two hill districts of Assam. Agriculture in the hill districts have their own problems like jhum cultivation, infrastructural weakness, low productivity etc. Intensive jhumming has accelerated soil erosion. But jhumming is a way of life for the hill people. Weaning them away from jhum cultivation is a great challenge before the State. Introduction of new schemes for production of cash crops, horticultural crops and settled agriculture in terrace lands will remain a priority in the coming decades. Infrastructural development and access to markets will be special areas of concern in development of hill agriculture. Exploitation of horticultural potentialities through introduction of suitable fruits and spices and their value addition will substantially contribute towards the economy of the hill people.

4.4 EMPLOYMENT GENERATION :

In any poverty alleviation effort, employment generation is an important component to facilitate distribution of income and equity. The open unemployment situation of the State has acquired a serious proportion. The burgeoning labour force of the State is comprised of relatively young people in the age group of 15 – 35 years. The employment situation can be tackled to a great extent by taking actions in two fronts, viz. service sector and farming.

SERVICE SECTOR EMPLOYMENT :

There is wide scope for creation of employment in the field of agro-service. Rural youths can take up employment by setting up inputs supplying outlets for seeds, fertilizers and pesticides combining with it custom hiring facilities of agricultural machineries. There are 2,486 Gaon Panchayats in Assam. Initially one Agro-Service Centre can be established in each Gaon Panchayat selecting youth having either I.T.I. qualification or knowledge of machineries. Each Agro-Service Centre will have at least one tractor with all accessories and trailer along with some other minor machineries and implements for custom hiring. The Centre will also have seeds, fertilizers and pesticides etc. for sale.

In the horticulture sector and marketing there will be vast employment potentialities in various services.

FARM SECTOR EMPLOYMENT :

The youths between the age group of 18 and 35 years form the bulk of State's labour force who are subject to rampant open unemployment. Formal educational system in the State cannot prepare them for jobs in modern sector employment nor self-employment scope in different enterprises is wide due to limited economic infrastructure. Employment avenue in government sector is also shrinking fast due to attempts at curbing government expenditure. As a result vast human capital remains unutilized for welfare of the State. The outcome of it all is that youths lose self-esteem, which creates discontentment.

Agriculture and allied sector has the potentiality to absorb the rural youths in large scale for gainful employment. An educated youth belonging to a household having at least 10 bighas of land can be self-employed with support from the government. Ten bighas of land have been considered here as a unit of operation which will allow certain economies of scale. If educated youths can be attracted in a large scale to agriculture, it will hasten transition of subsistence agriculture to diversified and commercial agriculture. The youths of lower income families of rural areas tend to perform poorly in their educational pursuit due to their disadvantaged circumstances. Private costs of education are higher for them and they often leave institutions without completing. Poor rural youths cannot progress in the educational ladder. Therefore, poor rural youths deserve preference in the proposed agricultural development programme.

It is estimated that there are about 3 – 4 lakh households in the State having land holdings of 10 bighas or more. But youths of these households to be brought under this programme will be much less. In the premises of the above background, a scheme is formulated to extend support to the rural youths of the State for employment in agriculture.

MODALITIES OF IMPLEMENTATION OF THE SCHEME :

1. FORMATION OF AN AGENCY FOR IMPLEMENTATION OF THE SCHEME :

A Young Farmers' Development Agency (YFDA) will be formed in each district to administer and co-ordinate various activities of the scheme with the Department of Agriculture as Nodal Department.

OBJECTIVES : i. Increase of agricultural production and incomes of beneficiaries through adoption of appropriate technology.

- ii. To ease the unemployment problem of the State.
- iii. Transition of subsistence agriculture to diversified and commercial agriculture.
- iv. Mobilization of savings from beneficiaries for future investment in agriculture.
- v. To restore confidence in the minds of the youths to increase self-esteem.

2. **TARGET :** 10,000 Youths in phases.

3. **SELECTION CRITERIA :**

- A. ESSENTIAL :**
- a. Age between 18 – 35 years.
 - b. Does not occupy an office of profit.
 - c. Presently a practising farmer.
 - d. The family of the beneficiary possesses at least 10 bighas of unencumbered Myadi Land and must not be scattered in more than two parcels. Land should be hazard free.
 - e. Successfully completed at least primary schooling.
 - f. Desirous of becoming a life long farmer.
 - g. Must possess sound health.
 - h. Must belong to financially weaker section of society.
- B. DESIRABLE :**
- a. Self or parent is a member of local Pathar Parichalana Samitee (FMC).
 - b. Proven record of interest in agriculture.
 - c. Knowledge of modern agriculture.
 - d. Higher educational qualification.
 - e. Must be inclined to serve society.

4. **SELECTION PROCEDURE :**

Applications will be invited through newspapers for selection of beneficiaries. The candidates will be selected through an interview and verification of required information under the supervision of Deputy Commissioner.

Selected candidates will be trained under a three months orientation course. On successful completion of the course and execution of a declaration to the effect that he / she will remain a life long farmer; the candidate will be registered under Agriculture Department as a beneficiary of the scheme.

5. **TYPES OF ASSISTANCE :**

- i. Each beneficiary will be given a monthly incentive of Rs. 200.00, which will be deposited in a Provident Fund specially created for the purpose. Beneficiaries will be allowed to temporarily withdraw money to meet contingent situation.
- ii. On attainment of 60 years of age each beneficiary will be given monthly pension. A Corpus Fund will have to be created for this purpose through a legislation.
- iii. Beneficiaries will be regularly provided with new technological information for improving productivity.
- iv. Beneficiaries will be provided with inputs from usual schemes to the extent possible.

6. FINANCIAL INVOLVEMENT :

a.	Monthly incentive for 10,000 beneficiaries @ Rs. 200.00	Rs. 2.40 Crores
b.	Subsidy to the extent of Rs. 70,000.00 for purchase of a Power Tiller. (Total No. 10,000)	Rs. 70.00 Crores
c.	Computers and Office Stationeries	Rs. 0.30 Crores
d.	Training expenditure for 10,000 beneficiaries	Rs. 2.00 Crores
e.	Capital and contingent expenditure	Rs. 3.30 Crores

		Rs. 78.00 Crores

7. MONITORING :

Success of this scheme will depend on constant monitoring of activities of the beneficiaries. Nature and quantum of information to be handled are too complex and numerous for manual operation. Therefore, computers are to be used in each district for maintaining basic information maintaining basic information of beneficiaries, their current performance, production level of agricultural commodities and disbursement position of monthly incentives etc.

4.5 PEOPLE'S PARTICIPATION :

People's participation is absolutely necessary for successful implementation of any programme. With this end in view the Agriculture Department, Assam started organizing a new social institution at grass root level called Field Management Committees (FMCs) or Pathar Parichalana Samittees (PPS) from last 10 years. This is a farmers' voluntary body consisting of 40 to 60 members living near a big cultivated field. There are about 25,000 FMCs in Assam at present at village level having coordination bodies at Block, District and at State level. They carry out mainly the functions of motivation and liaison with the Agriculture Department. The power and effectiveness of FMCs were demonstrated recently through successful installation of one lakh STW by Agriculture Department within a record time. The speed of implementation of the STW programme has hardly any parallel in India.

The tremendous social energy created by the FMCs must now be channelised for higher goals. In the present scenario of state agriculture, it is observed that an element of self-help must be introduced to tide over financial and marketing problems, the two weak links in the entire agriculture production chain.

FINANCIAL RESOURCE MOBILISATION :

The present financial resources of the state government are stated to be in bad shape. Yet the Govt. went all out to manage funds from national, international financial institutions for modernization of State's agriculture. This cannot go on endlessly without putting severe strain on states finances. Therefore, ways and means must be found out to mobilize resources internally. The FMCs can be persuaded to start saving on monthly basis which is in official parlance called Self Help Groups (SHGs). Total money saved at the rate of a monthly saving of a paltry sum of Rs. 30.00 by each member, with an average membership of 50 for each FMC can turn out to be substantial amount of fund for the entire State where there are about 25,000 FMCs.

Monthly saving	= Rs. 30.00 x 50 x 25,000
	= Rs. 3.75 crores
Annual saving	= Rs. 3.75 x 12 crores
	= Rs. 45 crores.

Saving will bring about a sense of confidence and self-respect into the minds of farmers. This will enhance their credit rating and will act as a symbol of self-reliance. Govt. may consider to partly support the managerial cost of this huge saving movement.

Finances mobilized in this manner will contribute to capital accumulation required urgently to increase output and income of farmers. Trade-off between current consumption and future investment will brighten the picture of Assam's agriculture.

The micro credit assumes great importance among the poor farmers in consideration of inability of Banks to serve the banking needs of large number of them. The FMCs can play a pro-active role in micro credit delivery to farmers.

A seminar on Green Revolution held at Guwahati on 19th and 20th June 2000 also placed on FMCs the role of an intermediary between Banks and farmers for smooth delivery of credit to farmers.

MARKETING AND COMMERCE :

Marketing of agricultural inputs and outputs is the weakest spot in Assam's agriculture. There are great scopes to create marketing services from storage and transport to value addition. These being costly, technological and managerial in nature, entrepreneurship has not developed in the State due to its aloofness and inherent shortage of it in the local people. But in the context of reforms in the Indian economy, our markets are to be integrated with the national and international markets, which will bring profit and competitiveness into our economy. Given the will necessary for the purpose, the FMCs can be corporatised and managed professionally on modern lines. This corporate body can function as the business wing of FMCs with a Brand image in the line of AMUL of Gujrat. The State can boast of many modern rice mills, oil mills in oilseed producing areas like Dhemaji, Lakhimpur and Majuli and fruit processing plants in Barak Valley, Hills and in the Plains through this corporate body. It can enter into fertilizer, seed, pesticide and agricultural machinery business both in the State and in the national level.

4.6 AGRICULTURAL INPUT :

Seed Requirement :

Seed is the basic input for increasing crop production and productivity. Therefore, maintenance of genetic purity through seed replacement is essential for stabilizing the yield levels. The existing seed replacement rate (SRR) in the State as a whole is very poor. Seed replacement would be enhanced substantially (annually) by 2025. Efforts should be made to produce adequate quantity of seeds of high yielding varieties suitable for rice under different rice eco-systems. Concentration of efforts in production of high yielding variety of seeds and hybrid seeds will be through registered seed growers. Required skill and technical guidance will be provided to such seed growers in production of quality seeds in the State. In order to extend the area under HYV / hybrid seeds, the farmers will be motivated to exchange their seed among themselves. Participation of private sector will also be encouraged for seed production in the State.

High yielding varieties recommended for various eco-systems :

- | | | |
|----|----------------------|--|
| 1. | Rainfed | Lakhimi, Bahadur, Kushal, Ranjit, Monohar Sali, Masuri, Moniram, Keteki Joha, Rangali and Bhogali. |
| 2. | Irrigated area | Rashi, Lachit, Chilarai, IR-36 and IR-64. |
| 3. | Post flood situation | Luit, Kapili and Kalinga |

ESTIMATED SEED REQUIREMENT

Area in lakh hectare
SRR (Seed replacement rate) in PC
SR (Seed replacement) in lakh MT

Crop	Present Replacement (in %)	2001-2002			2006-2007			2011-2012			2016-2017			2024-2025		
		Area	SRR	SR	Area	SRR	SR	Area	SRR	SR	Area	SRR	SR	Area	SRR	SR
1. Paddy	1.7	27	5	0.36	30	7.5	0.58	30	10	0.72	30	10	0.72	30	12	0.86
2. Pulse	5	1.6	12	0.04	2.1	20	0.10	2.55	30	0.18	2.8	40	0.27	3.2	50	0.38
3. Oilseed	2	4.03	13	.052	4.17	18	0.07	4.30	23	0.09	4.55	27	0.12	5	35	0.17

Hybrid rice seeds are to be introduced in phases in Summer rice which is not included above.

FERTILIZER :

Fertilizer in Assam is of special significance as the State has a very low level of consumption. Although the State has achieved some improvement in consumption of fertilizer during the current 9th Plan period yet consumption level is still low compared to national average. In view of the low level of consumption, it has been proposed to make all out efforts to substantially increase the consumption of fertilizer from the current level of 29.3 kg/hect to a level of 50 kg/hect by the end of 9th Plan and 70 kg/hect from 2005 - 2006 onwards. Focus of attention has also been given in feeding the soil by way of organic manure, bio-fertilizer and green manuring to maintain the soil health.

ESTIMATED REQUIREMENT OF FERTILIZER IN TERMS OF NUTRIENT

NPK in MT

Year	Season	Area	N	P ₂ O ₅	K ₂ O	Total
2001-2002	Kharif	25.00	49172	23986	23042	96200
	Rabi	12.00	45388	22142	21270	88800
Total :		37.00	94560	46128	44312	185000
2006-2007	Kharif	25.00	62500	31250	31250	125000
	Rabi	14.90	77150	38575	38575	154300
Total :		39.90	139650	69825	69825	279300
2011-2012	Kharif	25.00	62500	31250	31250	125000
	Rabi	16.80	83800	41900	41900	167600
Total :		41.80	146300	73150	73150	292600
2016-2017	Kharif	25.00	62500	31250	31250	125000
	Rabi	18.70	90450	45225	45225	180900
Total :		43.70	152950	76475	76475	305900
2024-2025	Kharif	25.00	62500	31250	31250	125000
	Rabi	19.65	93775	46890	46885	187550
Total :		44.65	156275	78140	78135	312550

4.7 IRRIGATION :

The State has proposed to install another one lakh Shallow Tube Well (STW) by exploiting the abundant ground water resource during 2000-01 immediately after completion of 70,000 Nos. of STWs currently being undertaken. It is expected that after completion of the proposed one lakh STWs, the assured irrigation coverage will stand at 7.27 lakh hectares during 2000-01 itself. The balance of 2.73 lakh hectares against the target of 10 lakh hectares would be covered in the succeeding year of 9th Plan by installation of STWs and by exploitation of surface water bodies abundantly available in Assam.

Keeping in sight the ground water reserve of the State, exploitation of surface water by way of Flow Irrigation Projects needs pointed attention. In addition construction of reservoirs in drainage systems will ease flood problem and create irrigation potential in the State.

4.8 AGRICULTURAL MECHANISATION :

Timeliness being the essence at every stage of an agricultural operation, dependence on the use of different types of machineries is inevitable. The increasing demand for utilisation of farm machineries also stems from the need to economic farming activities and minimise the cost component on labour, which is always on an increasing trend. Besides the availability to a labour force of other equally remunerative but with less drudgery activities than agriculture restrict the availability of work force especially at crucial stages during agricultural operations.

It is therefore necessary to introduce the use of different types of machineries whether it involves the use of tractors and power tillers for ploughing, or transplanters for sowing or a harvester after the standing crop has matured to even a mechanical drier to reach a desired level of moisture content. The wide range of machineries shall be the ultimate goal of the process of mechanisation in the State.

Mechanisation as a mode of practice in agriculture by the use of tractors and power tillers has been dotting the agriculture scenario in the State for many years now, although in a limited scale. Hitherto with the majority of the farmers cultivating only one crop in a year, a farmer has plenty of time in hand, and the need to rush through after harvest of one crop never occupied the mind of the farmer. It has only with the recent increase in coverage of area under irrigation, mainly through Shallow Tube Wells, that a farmer has now decided to cash in on the facility available with him and take to cultivation of an additional second or third crop, especially Rabi. There has been a perceptible increase in the area under

double crop in the State from 26 percent at the end of VIII Plan to 33 percent in 1999-2000 and expected to increase further to 50 percent by the end of the IX Plan. And therefore mechanisation has become inevitable means to march ahead rapidly to usher agriculture revolution in the state.

Mechanisation includes the entire gamut of agricultural operation starting from ploughing the land by means of power tillers and tractors to transplantation of seedlings by means of a transplanter down to harvesting of a crop harvester. The post-harvest activities would also require proper storage and reduction of the moisture content up to a desired level. Therefore, this would invariably point towards preparation of vision plan which would provide thrust on key focus areas for creating the impetus towards mechanized agriculture in the state.

In the IX plan (1997-2002) for mechanization of agriculture in the state, the following will be key focus areas.

The use of improved and efficient implements, pre and post harvest equipments and machineries by way of support programme and also emphasis will be provided for increasing availability of mechanical power in pre & post harvest agricultural operations. Agro-processing activities relating to perishable and durable agricultural produce in the rural areas will be encouraged for establishment of small-scale units (SSI) keeping export promotion in view. Farmers would be motivated for scientific storage of grains through metallic Bins etc. for grains saving.

Special emphasis would be laid on strengthening Agricultural Information Network to provide Agro-meteorological database for weather monitoring, forecasting, surveillance early warning system for natural calamities for efficient crop husbandry with mechanisation.

Augmentation of employment and income generation would be encourage through agricultural equipments and machineries servicing centres, Agro-processing industries consultancy, contract jobs, spare parts shops etc. in the rural areas of the state.

In the X PLAN (2002-2007), the key areas in the plan will be as follows :

Emphasis on increasing availability of mechanical power for pre & post harvest agricultural operations through Tractor, Power tillers, self propelled type Rice Transplanter, Tractor drawn modular Planter, self propelled type Reaper, power Paddy Thresher, Multiple crop Thresher, Harvester combine, Mechanical grain Driver etc. and to encourage post harvest processing through machines like Dal Processor, Oil Expeller, Potato peeler. Scientific storage through high metallic Bins and development of Cold chains for perishable agricultural commodities in the state will be given high priority. Promotion of servicing centers for agricultural machineries and equipments and also custom hire service with avenues for employment generation will be emphasized along with strengthening agricultural database management for efficient crop husbandry through machanization under Integrated Intensive Farming System. The strengthening R&D activities on location specific problems of relevant mechanization, training and development need of farmers specially of rural youth in agricultural mechanization will be given high preference.

In the XI PLAN : (2007-2012) key focus areas will include the following :

Emphasis on increasing availability of mechanical power for pre and post harvest agricultural operations through Tractor and other equipments and machineries for different crops including spices and horticultural crops and encouraging post harvest processing through machines like Dal Processor, Oil Expeller and other equipments for value addition keeping in view export promotion of agro-products. The scientific storage of grains through silos and metallic high bins for grain saving and also expansion of Cold chains for perishable agricultural produce will be prioritized. Expansion of agro-processing and agro based activities relating to perishables and durables in rural areas with export orientation, upgrading and strengthening Agricultural Information Network for efficient crop husbandry through advance planning, and strengthening R&D activities for technology up-gradation on the basis of location specific needs for agricultural equipments and machineries will be the prime consideration. The training and development need of farmers especially of rural youth on mechanization will invite special attention in key focus areas.

The XII PLAN : (2012-2017) will cover the following core areas in the planning on mechanization :

The increasing availability of mechanical power for pre and post harvest agricultural operations through Tractor and other equipments and machineries for different crops including spices, floriculture, medicinal and aromatic plants including other horticultural crops will be given emphasis. Encouragement of post harvest processing of different crops and also value addition for export promotion, scientific storage of grains and agricultural produce through silos, metallic high bins and other innovative structures for reduction of loss in saving would be emphasized. The upgrading and strengthening of Agricultural Information Network and database management for sustainable crop husbandry through mechanization with advance planning will be given due consideration. The strengthening R&D activities for technology up-gradation on location specific needs and also training and development needs for farmers specially of rural youth on farm machineries equipments will also be emphasized during the plan period.

The XIII Plan (2017-2022), shall concentrate primarily on the following key areas :

Stress will be given on increasing availability of mechanical and electrical power through diversified use of pre and post harvest agricultural machineries and equipments and also for encouraging post harvest diversified processing with value addition for export market through innovative equipments and machineries. Continuous up-gradation and strengthening of Agricultural Information System through *Net* for efficient crop and basic resource management. Opening up of employment avenues through ancillary units related to agricultural equipments and machinery manufacturing, servicing centres, consultancy and other jobs would be given due consideration. The strengthening of R&D facilities with support from knowledge centres and also development of patent for indigenous equipments and machineries under farm mechanisation will be given emphasis. The training and development need of knowledge workers as well as farmers for updating information on innovative technology and farm products would be the prime consideration for the plan period.

In the XIV Plan (2022-2027), the key areas in the perspective plan shall include the following :

Emphasis shall be given on increasing availability of farm power through improved models of tractors and other pre and post harvest equipments and machineries for different crops and to encourage diversified agricultural processing and value addition of product for export market at rural areas. Strengthening of capacity building for farm power infrastructure at FMC levels and sophisticated pre and post harvest equipments and machineries for horticultural crops, spices, medicinal and aromatic plants and other cash crops will be given preference. The strengthening of Information Centres on agriculture for farmers access through *Net* technology for efficient crop husbandry through satellite derived information from **INSAT** will be a key focus area of the plan. The promotion of database for entrepreneurship development for S.S.I. Units relating to agricultural equipments and machineries, consultancy, custom hire & servicing centres would be given priority during the plan. The strengthening R&D activities in knowledge centres for farm equipments and machineries for customized products, patent and innovative research on mechanisation will also be given due consideration. The training and development opportunity to knowledge workers as well as farmers at village level for updating information will be given high priority.

4.9 TRAINING :

Training is most important tool to improve effectiveness of people in their work place. In view of the importance attached to agriculture by Assam Govt. heavy financial investment planned by it and the necessity of absorbing a large young labour force in agriculture, training has assumed special importance. During the next two decades there will be great changes in demand of commodities, production and processing technologies, production relations and institutions. To cope with the changes, design of training and its delivery methods will have to be constantly updated. Solutions of today will be fast obsolescent. Therefore, a proactive training policy will be necessary to take care of necessities of changing times.

There are many traditional and new areas of learning for vast agricultural activities proposed during the next two decades for persons directly engaged in farming, persons directly engaged in off-farm activities and members of social institutions like Field Management Committees (FMCs).

There will be many new areas of skill development to sustain agricultural production in the coming decades. Some of the new learning areas are - cultivation of hybrid rice, pulse and oilseeds as intercrop and main crop throughout the year, new spices and condiments, floriculture, economics of water use, environmentally viable production methods, production for exports, agro-forestry, use of bio-control agents and biotech seeds, tissue culture plants, use and maintenance of new generation farm machineries and packaging etc.

Moreover, knowledge and skills of management like leadership, planning, communication, information systems use, business planning, budgeting, record keeping and market information use are important areas of learning. The extension machinery will also require training in the areas of precision farming, phytosanitation, food sanitation, eco-friendly farming system and general management.

In the light of the above, training infrastructures have been built in the state for middle aged and young farmers by the Directorate of Agriculture. A farmers' training programme for young farmers are to be formulated in every training center. Establishment of farmers' training schools were started in Karnataka in mid thirties and these are functioning till the present. Besides there are training centers set up for young farmers with international funding. Assam can also propose to set up training centers in each subdivision of the state in the near future.

Farmers' needs to be looked from three angles i.e., farm, farmers and farming system. The farmers need to decide on the output requirements and input needs for the resources (land, water and capital). The farming system involves the technology and package of practices for sustainable agricultural development. It is necessary to orient him on agro-climatic planning approach for his acceptability towards mixed farming strategies.

4.10 INFRASTRUCTURE :

1. ROADS :

Due to either absence or poor shape of rural roads, agricultural commodities in the interior villages are sold at non-remunerative prices or get wasted. Endeavours made so far in Assam to improve rural roads and bridges through the World Bank (ARIASP) and NABARD programmes as well as through District Rural Development Agencies have yielded good results. The scope of such works must be widened now to support the modernization of Assam's agriculture.

2. TRANSPORT AND STORAGE :

Marketing weaknesses have prevented stability in production and kept productivity of crops low. Improved storage, transport, market yards and market conditions will automatically reduce the gap of the price spread. Government support in transport particularly to provide refrigerate container trucks / vans for perishable commodities like flowers, fruits and vegetables will benefit the farmers immensely.

The importance of storage is hardly seen or realized. Very high degree of loss of cereals and grains occurs during storage which ranges between 20 to 50% loss. The importance of storage increases with the increase in production as seen in Assam at present. Most modern community storage structures like silo bins and for rural households ordinary metallic bins are required to be made available, in easy term. The storage space of Central and State Warehousing Corporations are also to be augmented immediately. This will help in reducing storage loss. Assam at present suffers badly from the dearth of cold storage space. Urgent steps is required to be taken to establish cold storage chain in the State in a time bound manner to reduce loss of perishable agricultural commodities. A new thinking for formation of central auction markets as well as improvement of rural markets are required to be taken up.

3. POST HARVEST MANAGEMENT :

Post harvest management of commodities though related to storage as mentioned above, there is specific areas to be attended to due to high rainfall in Assam. Assam has decided to go in a large scale for summer rice production. Harvesting of summer rice coincides with rainfall. Threshing and drying of summer rice during rainfall are serious problems faced by farmers. Therefore, in areas of summer rice production, drying yards and storage structure are to be constructed in villages to prevent summer rice loss.

4. AGRO-PROCESSING :

There is vast scope of value adding agro-processing industries in the State. With great surpluses in rice in the near future, establishment of modern rice mills will give a boost for export promotion. Besides, dal processing and oil extraction are two other fields which can benefit the State.

Considering the abundant production of fruits, vegetables and spices in the State, there are vast economic possibilities in the agro-processing sector. This has been discussed under Horticulture.

5. BANKING :

The banking infrastructure is poor in Assam. Banking must play a complementary role in modernising our agriculture. The present credit : deposit ratio of banks in Assam is low, i.e. about 30%. This has to be increased to much higher level in the future. Besides banking, other service infrastructures like insurance and marketing are also poor. Serious efforts are needed to activate the banking sector to participate in agricultural development of the State.

6. HI-TECH AGRICULTURE :

In the perspective of next quarter century, Hi-tech agriculture has a dominant role to play like production of horticultural and floricultural planting materials together with production of genetically modified seeds of crops to cut down the cost of insecticides and to increase productivity. Moreover, Hi-tech agriculture will be essential to promote exports and increase earning of the State. A high level State Committee or Authority may be formed to give direction to agricultural research including bio-technology in the State which will also oversee infrastructure development and entrepreneurship motivation activities.

4.11 TRADE WITH NEIGHBOURING COUNTRIES :

Assam is going to emerge in the near future due to its strategic location as door way to the South-East Asian markets. There is great scope for export of agricultural produce of the State / Region to these South-East Asian countries. Opening of more border trade, Stillwell high way and conversion of the L.G.B. Airport to an International Airport will stimulate trade with S.E. Asian countries. In order to meet international standard for export of our produce, phytosanitation and food sanitation will be new dimensions of marketing.

4.12 W.T.O. REGIME AND ITS IMPLICATION FOR ASSAM :

The World Trade Organization (WTO) evolved through Uruguay Rounds of Negotiations of General Agreement of Tariffs and Trade (GATT) and was born on April 15, 1994 in the Ministerial Meeting held at Marrakesh in Morocco. India has ratified the WTO Act and is bound to give effect to the Agreement on Trade Related Aspects of Intellectual Property Rights (TRIPS). In accordance with the provisions of TRIPS, Govt. of India introduced a Bill in Parliament on 14th December, 1999 called "THE PROTECTION OF PLANT VARIETIES AND FARMERS' RIGHTS BILL 1999". This Bill was sent to a Joint Parliamentary Committee and has been returned to the House in April. The Joint Parliamentary Committee visited Assam on 09.02.2000, held discussion with farmers, scientists, planners and government officials to elicit their opinion. The main purpose of the Bill is to provide for the establishment of an Authority to give an effective system for protection of the rights of the plant breeders and farmers, and to encourage the development of new varieties of plants. There are certain reservations about this Bill, which are discussed below.

1. Although the Bill has dealt in considerable detail about protecting the rights of plant breeders, the farmers' interests were confined to a single paragraph. The Section 31 states that "Nothing contained in this Act shall effect the right of a farmer to save, use, exchange, share or sell his farm produce of a variety protected under this Act :

Provided that a farmer shall not be entitled for such right in case where the sale is for the purpose of reproduction under a commercial marketing arrangement".

The above section does not allow economic gains to our farmers. Even if government wants seed production of a variety protected under this Act through a seed village programme or certified seed production through Field Management Committees (FMCs), it will not be possible under this Act. Unlike west, our society is cohesive and farmers' activities are community orientated. Therefore, there should be relaxation in this Act for community purposes.

2. The Bill has no provision to protect traditional knowledge of indigenous and tribal societies living in small areas. These societies have built knowledge systems around medicinal properties of herbs and plants. There is scope of multinational pharmaceutical companies pirating and patenting their knowledge with no benefit reaching to those communities which devoted centuries to build such knowledge.
3. New varieties are registered if they fulfill the criteria of "novelty, stability, uniformity and distinctiveness". Traditional societies also have bred many varieties through selection over ages. But they cannot fulfill the requirements of registration.
4. The North Eastern States are very rich in biodiversity and tribal communities. Law should protect their rights to prevent piracy of their seeds, plants, bio-diversity which have been preserved by the farming and tribal communities. It is not known how the Biodiversity Bill to be introduced in the monsoon session of Parliament can protect our biodiversity.
5. On the clause of benefit sharing, the Bill stipulates that claimants to benefit will have to prove their contribution to creation of knowledge of a new variety. Farmers lack the capacity to do it. The Authority proposed under the Bill has no obligation to make its own inquiries to confirm whether anybody has pirated traditional knowledge or whether the claimant deserves benefits.
6. Patenting is allowed only on new knowledge or mutated genes under TRIPS. On the basis of it, India was able to challenge the neem patent given in the U.S.A. Such state interventions are possible for products like turmeric and basmati rice used over wide regions of the country. But tribal and farming communities of small areas who have developed knowledge cannot protect their rights.
7. The multinational companies have entered the Indian seed market in large scale. It is possible for them to produce new plant varieties with state-of-the-art technology in genetic engineering. The possible impact of genetically engineered seeds on the environment and human and animal health is not known. However, the Bill has provision to exclude registration of a variety under Article 29 where prevention of commercial exploitation of such variety is necessary to protect public order or public morality or human, animal, plant life and health or serious prejudice to environment. The proposed Bill does not put the onus on the breeder to prove the

environment and health related impact of the new seed. Therefore, multinational seed companies will have free play to act without any concern for health and environment. The pertinent question is who will do it.

8. There are certain rice varieties unique to Assam like perfumed and glutinous varieties. Besides, Assam is extremely rich in biodiversity of citrus wealth. The plant breeders of Assam should make endeavours to register more productive varieties of rice and citrus under the Plant Varieties Bill. Then these breeders become automatically eligible to register their varieties in the convention countries with benefits coming on the heels of such registration.

Besides the implications of TRIPS related matters discussed above, there is another important instrument of WTO called the Agreement on Agriculture (AOA). The AOA favours transnational agribusinesses. The multinational seed companies will have access over agricultural resources and markets. The importance of providing necessary flexibility for the adoption of domestic policies by central and state governments has been impeded which forces import of food and markets being flooded by multinational seed companies.

4.13 INFORMATION TECHNOLOGY :

In keeping with the requirements of the present information age to store, process and transmit information, Information Technology (IT) which is a mix of computers and telecommunication has developed as a potent tool for fast socio-economic progress. Although Assam missed the boat of technological revolution in the past, she cannot afford to miss the present one. IT for agricultural sector cannot be seen separately from IT for the people. The fields where IT inputs will be highly productive in agriculture are given below.

1. ADMINISTRATION :

Agriculture department has a very high degree of public interface. In order to make this department effective and transparent, use of IT has become very important. This can be achieved by taking advantage of existing government telecom network to link all Block, Sub-divisional, District, Agricultural Offices with the Zonal Offices, the Directorate and the Secretariat and Agricultural Research and Training Institutes within the next five years.

2. FIELD MANAGEMENT COMMITTEES :

The FMCs have been playing a vital role in Assam's agriculture. This institution has been projected as the hub of the future agricultural development activities in the state. In order to empower these 25000 and odd FMCs in the state, a scheme may be proposed to network the FMCs which will be owned and managed by themselves. In this connection, the advice of Dr. M.S. Swaminathan, UNESCO Chair in Ecotechnology and Chairman, M.S. Swaminathan Research Foundation, Chennai in the Seminar on Green Revolution held at Guwahati on 19 and 20 June, 2000 that user-controlled, managed Knowledge Centres based on IT should be established throughout the state. These Knowledge Centres may be established in every Gaon Panchayat and the FMCs may be linked to these knowledge Centres.

3. INFORMATION ON UNEMPLOYMENT :

IT may be used to ease the unemployment position in rural areas by training and providing job in the IT sector. IT may also be used to monitor the knowledge and skill levels of the unemployed and in design of training to make them fit for gainful employment.

4. INFORMATION ON PRODUCTION TECHNOLOGY AND MARKET :

IT may be used to provide access to farmers to technological information on production methods and marketing of inputs and produces.



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Indian Council of Agricultural Research
Krishi Bhawan, New Delhi



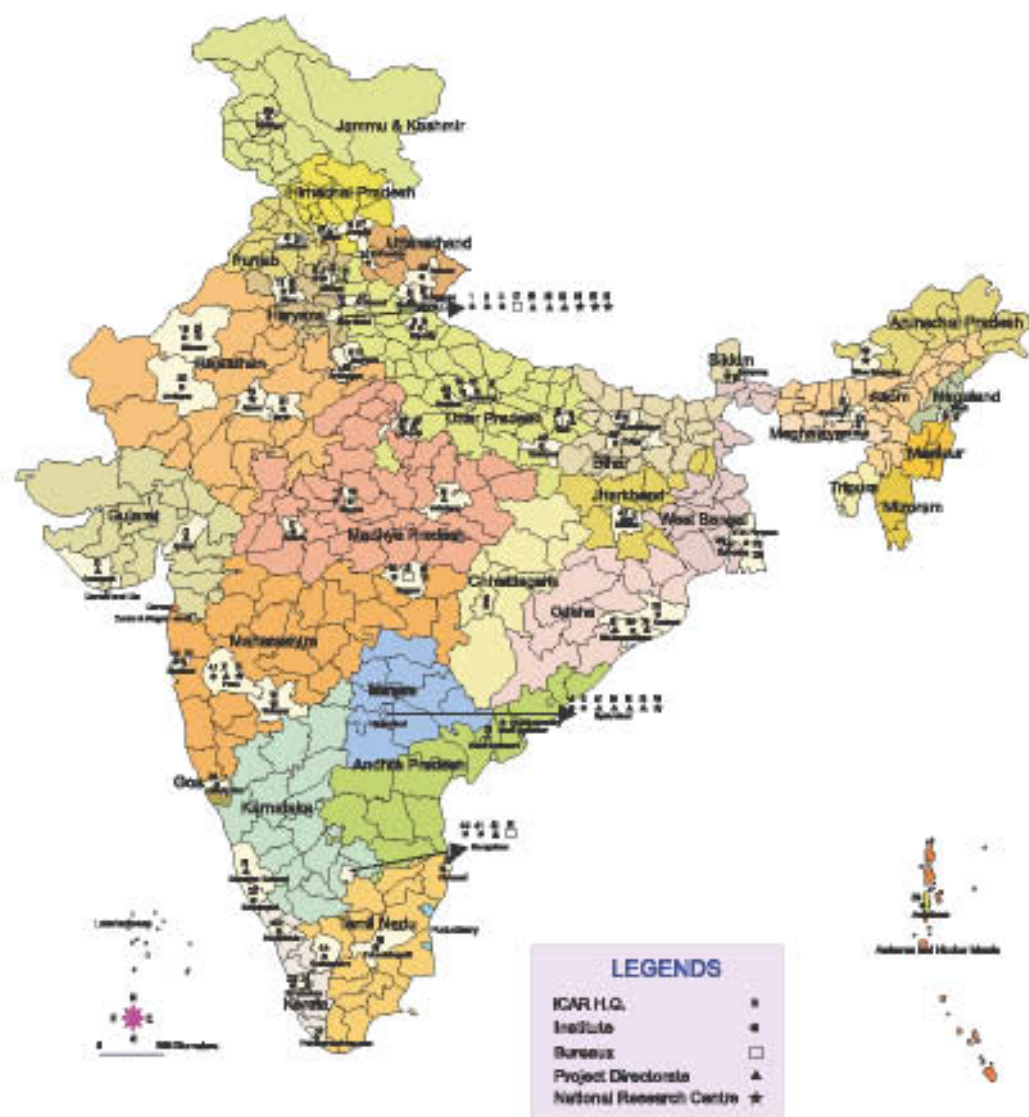
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Vision 2050



Indian Council of Agricultural Research
Dr Rajendra Prasad Road, Krishi Bhavan
New Delhi 110 001, India

www.icar.org.in

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संदेश



भारतीय सभ्यता कृषि विकास की एक आधार रही है और आज भी हमारे देश में एक सुदृढ़ कृषि व्यवस्था मौजूद है जिसका राष्ट्रीय सकल घरेलू उत्पाद और रोजगार में प्रमुख योगदान है। ग्रामीण युवाओं का बड़े पैमाने पर, विशेष रूप से शहरी क्षेत्रों में प्रवास होने के बावजूद, देश की लगभग दो-तिहाई आबादी के लिए आजीविका के साधन के रूप में, प्रत्यक्ष या अप्रत्यक्ष, कृषि की भूमिका में कोई बदलाव होने की उम्मीद नहीं की जाती है। अतः खाद्य, पोषण, पर्यावरण, आजीविका सुरक्षा के लिए तथा समावेशी विकास हासिल करने के लिए कृषि क्षेत्र में स्थायी विकास बहुत जरूरी है।

पिछले 50 वर्षों के दौरान हमारे कृषि अनुसंधान द्वारा सृजित की गई प्रौद्योगिकियों से भारतीय कृषि में बदलाव आया है। तथापि, भौतिक रूप से (मृदा, जल, जलवायु), बायोलोजिकल रूप से (जैव विविधता, हॉस्ट-परजीवी संबंध), अनुसंधान एवं शिक्षा में बदलाव के चलते तथा सूचना, ज्ञान और नीति एवं निवेश (जो कृषि उत्पादन को प्रभावित करने वाले कारक हैं) आज भी एक चुनौती बने हुए हैं। उत्पादन के परिवेश में बदलाव हमेशा ही होते आए हैं, परन्तु जिस गति से यह हो रहे हैं, वह एक चिंता का विषय है जो उपयुक्त प्रौद्योगिकी विकल्पों के आधार पर कृषि प्रणाली को और अधिक मजबूत करने की मांग करते हैं।

पिछली प्रवृत्तियों से सबक लेते हुए हम निश्चित रूप से भावी बेहतर कृषि परिदृश्य की कल्पना कर सकते हैं, जिसके लिए हमें विभिन्न तकनीकों और आकलनों के मॉडलों का उपयोग करना होगा तथा भविष्य के लिए एक ब्लूप्रिंट तैयार करना होगा। इसमें कोई संदेह नहीं है कि विज्ञान, प्रौद्योगिकी, सूचना, ज्ञान-जानकारी, सक्षम मानव संसाधन और निवेशों का बढ़ता प्रयोग भावी वृद्धि और विकास के प्रमुख निर्धारक होंगे।

इस संदर्भ में, भारतीय कृषि अनुसंधान परिषद के संस्थानों के लिए विजन-2050 की रूपरेखा तैयार की गई है। यह आशा की जाती है कि वर्तमान और उभरते परिदृश्य का बेहतर रूप से किया गया मूल्यांकन, मौजूदा नए अवसर और कृषि क्षेत्र की स्थायी वृद्धि और विकास के लिए आगामी दशकों हेतु प्रासंगिक अनुसंधान संबंधी मुद्दे तथा कार्यनीतिक फ्रेमवर्क काफी उपयोगी साबित होंगे।

(राधा मोहन सिंह)

(राधा मोहन सिंह)

केन्द्रीय कृषि मंत्री, भारत सरकार

Preface

The Indian Council of Agricultural Research is an apex research organization of the country with a high standing amongst international agricultural research institutions. Since its inception in 1930, the Council has been spearheading agricultural research, education and extension activities for productivity enhancement and diversification of Indian agriculture.

The world as a whole is undergoing several transformative changes. Growing population, changing lifestyles, expanding urbanization and accelerated climate changes are creating new challenges for the national agricultural research system. Whereas in the past, the challenge was to supply adequate food, but now it is to provide adequate nutrients to promote health; and in the future, the challenge would be to provide optimal nutrients based on individual's genetic profile. Fortunately, along with challenges, the developments in science are creating new avenues for tackling the challenges. The Indian Council of Agricultural Research (ICAR) and the National Agricultural Research and Education System at large, are determined to harness the advances of science for the welfare of society. The Council is committed to transform itself into an organization engaged fully with the farmers, industry, entrepreneurs and consumers at large.

To keep pace with the changing environment, the ICAR has been updating its visions and strategies from time to time. The first systematic effort to envision the challenges and opportunities, and formulate its own strategy was undertaken in the last year of the 20th century by preparing 'Vision 2020 document'. The next attempt was after five years by bringing out the 'Perspective Plan' and the 'ICAR Vision 2030', coinciding with XI plan. The present document, 'ICAR Vision 2050', provides the strategic framework for innovation-led inclusive and sustainable agricultural growth in the country.

I would like to express my gratitude to the Hon'ble Union Minister of Agriculture for his invaluable guidance in preparing the ICAR Vision 2050. I am grateful to the Hon'ble Union Minister of State for Agriculture for his keen interest in this initiative. We are thankful to all the Members of the Governing Body of the ICAR for their valuable suggestions in finalizing this document.

I am sure that ICAR Vision 2050 would stir a new thinking in the researchers to harness science, in the policymakers to develop policies for sustainable development of agriculture to provide food, income and livelihood, and in the consumers an urge to tailor their lifestyle, keeping in view the planetary boundaries of the Earth's resource system.



(S. Ayyappan)

Secretary, Department of Agricultural Research & Education (DARE)
and Director-General, Indian Council of Agricultural Research (ICAR)
Krishi Bhavan, Dr Rajendra Prasad Road
New Delhi 110 001

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Agriculture Scenario, Challenges and Opportunities

Humanity faces numerous challenges and most of them are linked to food security of the exploding population which is expected to cross the 9 billion mark by 2050. Globally, the contribution of agriculture to Gross Domestic Product (GDP) is declining, but agriculture still supports the livelihood of more than 2.6 billion people in the world, living mostly in rural areas. In low-income countries of Africa and Asia, potential for non-farm employment is limited, viability of subsistence agriculture of smallholders is under stress, and inequity is widening. The extraordinary progress in food production in the past half century was also accompanied by unsustainable levels of consumption in some countries and hunger and malnutrition in some other countries, largely in South Asia and Africa, leading to poor health, reduced earning capacity and degradation of environment. The fast emerging phenomenon of climate change is putting more pressure on the natural resources we depend on, and the safe space for operation within the planetary boundaries of earth's resource system is shrinking. It is time to rethink on how food is cultivated, processed, shared and consumed. There is a growing concern for development that ensures human happiness as embedded in the UN Millennium Development Goals and is the first principle of the 1992 Rio Declaration, which states, "*Human beings are at the centre of concern for sustainable development*" (UN, 1992).

Table 1 Status of selected global parameters

Parameter	Value	Parameter	Value
Population (2011)	7 billion	Area of agricultural land (2009)	4.9 billion hectares
Undernourished people (2010)	0.9 billion	Growth in agricultural production (1997–2007)	2.2% per year
People living on less than USD 1.25 per day	1.4 billion	Losses due to extreme climatological events	USD 11.4 billion
People living in dryland areas (2007)	2 billion	Food lost or wasted annually	1.3 billion tonnes

Source: Beddington et al. (2011).

A profound change in the global food and agriculture system will have to be incorporated, if we are to feed today's 925 million hungry people,

of which 230 million live in India. About 2 billion people are expected to be added to this category by 2050, mostly in developing countries, if corrective measures are not applied. On the brighter side, food security and sustainable agriculture are now on top of the global development agenda (UN, 2012). Sustainable agriculture is central to everlasting development and the judicious application of science and technology can help to achieve “zero hunger” goal even much before 2050.

Indian Agricultural Economy

Currently, the agriculture sector contributes only about 13.7 per cent to the national GDP (compared to 30% in 1990), though agricultural production in 2013 was at all-time high of 264 million tonnes. High dependence on agriculture for livelihood support (more than 50% population) is the main factor responsible for the high share of poverty in rural population. The agricultural economy that grew at an average rate of 3.7 per cent per year during 2007-2012, is projected to grow annually at 4.0 per cent during 2012-2017, with growth in national GDP estimated at 8 per cent. However, it must be remembered that the GDP estimates do not take into account the costs of environmental degradation. Agriculture places considerable load on environment in the process of production of goods and services. A template for estimating green national accounts, which would measure national production while allowing for the negative effects on national resources, is contemplated.

Further, India is predominantly a small farm agriculture. According to 2010-11 Agriculture Census, the total number of operational holdings was 138.35 million with average size of 1.15 ha. Of the total holdings, 85 per cent are in marginal and small farm categories of less than 2 ha (GOI, 2014). These small farms, though operating only on 44 per cent of land under cultivation, are the main providers of food and nutritional security to the nation, but have limited access to technology, inputs, credit, capital and markets.

The current trends indicate that by 2035 the growth rate of rural population will turn negative, leading to increase in the size of cultivation units, tightening of rural markets and increased mechanization of agriculture. By 2050, very few well-trained and farmer-cum-entrepreneurs will produce food. Their production systems are likely to be more commercial-oriented, operating at high productivity levels in a highly mechanized environment. It is more likely that today's small holders will organize themselves into producer companies and will have more bargaining power. Therefore, the country's research and development agenda has to be re-oriented to develop and promote such technologies

that would raise agricultural income and generate employment opportunities in the agri-supply chain. Though being limits on land-based agriculture's capacity to provide decent livelihood, the right course for growth of rural economy would be to ensure increased off-farm opportunities in bio-industrial complexes within the village clusters.

Drivers of Food and Agriculture Systems

The evolution of food and agriculture systems is complex and their numerous drivers have been traditionally viewed from one of two perspectives: multidimensional sustainable development perspective (environmental, social, economic, technological and policy-related dimensions); or essentially economic food demand-supply perspective. The need is to integrate both these perspectives across the research agenda of ICAR and NARES.

Drivers of Food Demand

The exploding population, expanding urbanization and rising incomes have raised a wide range of important issues linked to national food-security, including dietary preferences (higher demand for livestock products), consumption of more processed foods and crowding out of peri-urban agriculture which plays a significant role in the supply of perishable commodities (FAO, 2012). The population of India is projected to be 1.65 billion by 2050 with an average income of **₹ 401839/cap**, up from the level of **₹ 53331/cap** in 2010-11, with 50 per cent people residing in the urban areas (Table 2).

Table 2 Drivers of food demand and projections to 2050

Driver	2010-11	2050	Implications
Demand Side Challenges			
Population, million	1224.6	1650 (+24 %)	Changes in demography with shift towards urbanization and higher income will lead to changes in dietary composition with increased consumption of meat, dairy, egg and fish products having higher water and carbon foot prints
GDP at PPP (US\$ billion)	4786 (WB) *	43180 (PWC) **	
Average annual income (₹/cap)	53331***	401839***	
Income distribution	Inequitable	Inequity may widen	
Urbanization (%)	31	55	
Calorie requirement (kcal/cap) Source: (Vegetables; Animal)	2500 (92:8)	3000+ (84:16)	
Food price	Rising trend	Rising trend will continue	
Functional food	It is at the initial stage	Demand will grow	
Social concerns over new technologies	Very strong	Will decline with proper communication and counseling.	
Food safety	Poor	Will improve	

Source: *World Bank (2014), **PWC, (2014), ***NCAP (2013)

The size of population in the middle-income bracket will exceed 600 million (about 30% of the total population) by 2050. It is projected that by 2050, the calorie consumption will reach 3000 kcal/cap, with rise in the share of animal-based calories from the current level of 8 per cent to 16 per cent. This will increase the demand for food grains. In many ways, the Indian diets may slowly come closer to the diets in developed countries, necessitating production or import of food commodities (functional and special foods), consumed previously only in Western societies. Further, urban population being more vulnerable to increase in food prices and price spikes, it may adversely affect the food security scenario. Food wastages and losses also constitute an important driver of food demand. The estimates though vary, the wastages and losses of perishable commodities (fruits, vegetables, fish, meat and milk) can be put on average in 15-25 per cent range, while loss of non-perishable commodities may be anywhere between 5-15 per cent and this puts a considerable pressure on natural resources. By 2050, these losses are expected to be reduced significantly due to better use of technology and establishment of value chains.

Food safety is an integral part of food security, because unsafe food has significant economic costs in terms of health problems as well as cross border trade, which is hampered by inconsistent/poor food safety standards. The latent demand for food safety among urban India is set to grow with urbanization and will also increase acceptability in the export market.

There are various projections of increase in demand for food commodities in India. According to one scenario (Kumar, 2015), at 7 per cent growth rate in national GDP, though the demand for food grains will only grow by about 50 per cent, the rise in demands for fruits, vegetables and animal products will be more spectacular, the range being 100-300 per cent (Fig 1). Achieving such a high productivity will entail increase in total factor productivity (TFP). One-third of TFP must contribute to the agricultural growth. The food production density (kcal/km²/day) requirement in most parts of India will increase from 50,000-100,000 in 2005 to 100,000-500,000 by 2050, requiring a rise in foodgrain productivity from 25000/kcal/ha/day to about 46000 kcal/ha/day (Chaumet et al., 2009).

Supply Side Drivers

A general view is that close to 90 per cent of the growth in crop production is expected to come from higher yields (potential increase by biotechnology manipulations); and increased cropping intensity, which

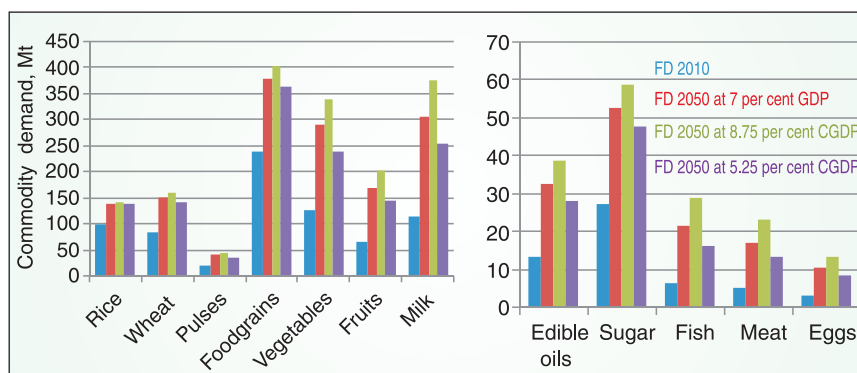


Fig. 1 Demand for various food commodities (FD) in 2010 and projections to 2050 at GDP growth scenarios of 7, 8.75 and 5.25 per cent
(Source: Personal communication from P. Kumar, 2015)

will mostly be possible in land facilitated with irrigation. It is expected that water-use efficiency would increase and groundwater withdrawals for irrigation would grow at a slower pace but still increase by almost 45 per cent by 2050. In medium-term, the economic growth is projected to continue at fairly sustained rates of 7-10 per cent. Due to food-feed-fuel conflict, the livestock rearing would be an intensive system with considerable reduction in numbers with simultaneous increase in productivity to meet the production targets. The supply side drivers can be grouped into three categories—environmental (e.g., climate, biodiversity, water resources etc.) economic (e.g. labour, energy, markets, investments etc.) and technology (e.g. biotechnology, nanotechnology, information technology, etc.) drivers.

Climate change: Climate change presents a major risk to long-term food security as it may have multidimensional debilitating effects on agriculture. The climatic impacts could lead to a dramatic scarcity of freshwater in the northern and peninsular regions of the country, fuelling the existing internal or interstate conflicts and heightening the competition among different users. Increasing seawater acidity and rising river water temperature will affect fish breeding, migration and harvest. Irrigated wheat and maize yields may decline by 5-10 per cent by 2050. Rain-fed agriculture, which covers 60 per cent of all the cultivated land in India, will be particularly hard hit. Except for the sates of Andhra Pradesh, Tamil Nadu and Karnataka, where yield of rain-fed rice is likely to go up by 10-15 per cent, rice yields will go down by 15-17 per cent in Punjab and Haryana and by 6-18 per cent in all other regions (CRIDA, 2013). In the longer-term, the production-depressing effects of climate change are likely to outweigh any production-boosting effects

under business as usual (BAU) scenario. But, agriculture is also in a position to offer solutions to climate change; it could be a strategic asset as a major provider of environmental services (sequestering carbon, managing watersheds, and preserving biodiversity) and could provide a way to increase farm income with the introduction of payment for environmental services. Livestock has always acted as insurance during environmental stresses and is more dependable than crops. But it would require technological innovations to make them adapt to climate extremes.

Declining/degrading land resource: India is a land-scarce country with only 0.13 ha land/cap (in 2010-11), which stands to get reduced to 0.09 ha/cap by 2050, simply due to increase in population. Further, land is getting polluted with toxic waste waters and there is a large-scale degradation due to water and air erosions. Annually, India is losing 0.8 Mt of N, 1.8 Mt of P and 26.3 Mt of K. Growing nutrient deficiency and declining factor productivity are the matters of great concern. Overall, the land balance for India is negative, i.e. land classified as not suitable, is being made productive through human intervention and the most productive land is being lost to the urban and industrial sectors. Much of the remaining land suffers from constraints such as ecological fragility, low fertility, toxicity, and would require high input-use and management skills, and prohibitively high investments.

Water scarcity and quality: Changes in the large-scale hydrological cycles due to climate change are the indicators of a fast emerging water-deficit country. As against the projected total water demand of 1498 billion cubic metres (BCM) (2030 WRG, 2009), the available supply is only 1121 BCM (CWC, 2013) and close to 50 per cent crop agriculture is likely to remain rainfed. Further, natural water systems, on becoming hydrologically water deficit, will lose their dilution capacity, leading to a higher concentration of chemicals and other pollutants. These factors would create numerous water management challenges, in both rivers and groundwater across the country. Groundwater being a major source of irrigation, the inbuilt water and energy nexus will have a big impact on energy requirement. Most northern and peninsular regions are already water-stressed and, are fast becoming water-scarce, both physically and economically. This will further aggravate the existing internal or interstate conflicts besides heightening competition among different users. According to a study by the International Food Policy Research Institute, 45 per cent of the total GDP (US\$ 63 trillion) will be at risk due to water stress by 2050 (Rosegrant *et al.*, 2005).

Biodiversity: Biodiversity of plants and livestock, which is so crucial for sustaining long-term productivity, is under threat. The rate of extinction is alarming, as only four crops provide about 60 per cent of global food, causing declines in genetic diversity among cultivated species. The genetically uniform systems are extremely vulnerable to external shocks under extreme weather conditions and emerging diseases and pathogens. Since 1900s, about 75 per cent of crop diversity has been lost from the farmers' fields. Similar is the case with livestock where the number of indigenous breeds with better adaptability, disease-resistance and feed-use efficiency is declining. Conservation and improvement of such breeds are foremost on the agenda of the Council.

Pandemic pest and diseases in animal and plant production: Movement of plant pests, animal diseases and invasive alien aquatic organisms does not recognize physical and political boundaries. Globalization of commodity and food trade has increased the bio-security risk, which threatens food security. It is estimated by FAO (2007) that 80 per cent of all wheat varieties planted in Asia and Africa, are susceptible to new virulent wheat fungus which spreads rapidly by wind. The spread of Severe Acute Respiratory Syndrome (SARS), blue tongue and avian flu in the livestock sector, and emergence of contagious human diseases from animals and the current epidemic of highly pathogenic avian influenza (HPAI), in densely-populated livestock-production areas around cities increase the production risks and present a major challenge to ensuring food security in the country.

To sum-up, environmental drivers like climate, water, land and biodiversity are extremely important for agricultural production, but are being degraded gradually making the food production systems increasingly vulnerable.

Economic Drivers

Food systems are vulnerable to macroeconomic trends such as stagnation, recession, commodity prices and exchange rates and therefore, to the terms of trade. The major issues are energy, labour and farm mechanization. Energy-use in agriculture for pumping water for farming is projected to increase at the rate of 1.78 per cent per year. The shifting paradigm of bioenergy, which creates new scarcities and threatens food supplies worldwide, may result in food insecurity. But, there is ample scope for harvesting solar energy for which countrywide efforts are underway. Though in absolute terms the energy consumption will go up, the energy-use intensity will have to go down. The increase in food prices may adversely affect food security. The share of labour

in agriculture will decline through “push” of labour productivity improvement and “pull” of labour demand from other sectors. With industrial growth and consequent depopulation of the rural areas, the farm labour availability will decrease, causing faster mechanization of farm operations, which in turn will create need for skilled labour for operation and maintenance of farm machinery. It is expected that there will be a big increase in public spending on innovation and technology generation and therefore a mechanism to monitor its effectiveness will have to be put in place. Further, following the global trend, the private investment in agriculture, and more specifically in food research, will considerably grow in India also.

Technology Drivers

Technology advancements in agriculture and food sector will be characterized by the increasing convergence across bio, nano and info technologies. Innovations in the industrial sector will strongly influence advanced technology in agriculture and food processing, bringing in robotics and automatization in the food and agricultural sector. Mechanization of farm operations through energy-efficient and environment-friendly devices will compensate for the growing shortage of farm labour. Developments in biotechnology and other frontier sciences are creating tremendous scope and opportunities for use of agricultural products and by-products in innovative and unconventional forms. The Council, through its Bureaus, is the custodian of huge *in-situ* and *ex-situ* collections of genetic stocks of crops, animals, fishes, insects and microbes. These stocks will be useful in genetic improvement of agricultural commodities through conventional techniques and we will increasingly resort to frontier scientific techniques. Information technology (IT) systems are the core component in the transition from breakthroughs in labs to field-scale implementation. A combination of systems-research tools relating to information technology, geographic information systems (GIS), global positioning systems (GPS), remote sensing (RS); and climate smart resource management technologies, which enable practising precision agronomics, have become available. These technologies will help optimize land use to maximize calorie production per hectare on a global scale. As reported in the 2011 Global Agricultural Productivity Report, the average GPS-induced yield gains were of 10 per cent with an average input savings of 15 per cent in the USA (Global Harvest Initiative, 2011).

Smart sensors and new delivery systems will help combat viruses and other crop pathogens. Though some products and methods based on

nanotechnologies, genetically modified organisms (GMOs), nutraceuticals and nutrigenomics have been introduced in the market, their impact on social organization environment, risks of pandemic diseases, and human health remain to be ascertained. Research on animal health also stands to benefit from the innovative developments in medical sciences. Further, liberalization in trade and global access will lead to greater movement of seeds/plants, planting materials and consequently, will enhance chances of introduction of exotic pests. An effective regulatory approval process for new technologies, which allows farmers to gain access to the latest technological advancements, will have to be put in place.

Markets and Value Chains

India's food processing sector is expanding in response to the changing demographics, availability of strong local and international brands, emerging modern retail chains and growing consumer demand for processed foods. The food processing sector accounted for 1.5 per cent of national GDP and was valued at US\$ 14 billion during 2013-14. It was envisaged that the level of processing of perishable agri-commodities would be increased from 6 per cent in 2010 to 20 per cent by 2015, enhancing value addition from 20 per cent to 35 per cent. The public policies are geared towards diversification to high-value agriculture — fruits, vegetables, dairy and poultry, etc. — which can generate higher incomes on small farms by establishing shorter value chains and linking farmers to markets. Agri-food marketing systems are likely to undergo a significant transformation towards demand-driven, vertically-coordinated systems, managed by the agribusiness and marketing firms. Integrating small farmers into the demand-driven supply chain through appropriate institutions and policies would be a challenge. A national agricultural commodity market is proposed to be established to increase access to markets.

Growing Research Partnership and Private Sector Investment

There is an encouraging trend of forging partnerships amongst NARES and international research institutions. Globally, the share of private investment in agricultural research has become significant, as it now accounts for about a third of the total research investments; and is still rising rapidly. In the recent past, the ICAR has also graduated from aid-based research funding support to a loan-based funding such as the World Bank supported programs like National Agricultural Technology Project and National Agricultural Innovation Project, conducted in the PPP mode and making it more accountable. It is expected that

in future research partnerships among public institutions, CGIAR institutes, foundations and private companies will grow, leading to a faster generation of technology and its dissemination to the stakeholders. Strict implementation of IPR and patent regime will create a favourable environment for a sizeable investment in research by the private sector companies.

Summing up, the productivity paradigm of the past 50 years will give way to a new paradigm of multidimensional approach to agricultural research with emphasis on environment, health, nutrition, consumer preferences and farmer livelihoods, in addition to productivity. The changes in the status of major supply side drivers by 2050 are summarized in Appendix I.



National Agricultural Research System and ICAR

The Indian National Agricultural Research System is one of the largest systems in the world in respect of human source engagement and infrastructure, and the Indian Council of Agricultural Research (ICAR) is an apex body of this system. The Council administratively is an autonomous organization under the Department of Agricultural Research and Education, Ministry of Agriculture, Government of India. The ICAR coordinates, guides and manages research, education and extension services in agriculture, including crops, horticulture, agroforestry, fisheries and animal sciences.

Mandate

- To plan, undertake, aid, promote and co-ordinate education, research and its application in agriculture, agroforestry, animal husbandry, fisheries, home science and allied sciences.
- To act as a clearing house of research and general information relating to agriculture, animal husbandry, home science and allied sciences, and fisheries through its publications and information system; and instituting and promoting transfer of technology programmes.
- To provide, undertake and promote consultancy services in the fields of education, research, training and dissemination of information in agriculture, agroforestry, animal husbandry, fisheries, home science and allied sciences.
- To look into the problems relating to broader areas of rural development concerning agriculture, including post-harvest technology by developing co-operative programmes with other organizations such as the Indian Council of Social Science Research, Council of Scientific and Industrial Research, Bhabha Atomic Research Centre and the universities.
- To undertake any other activity considered necessary to attain the objectives of the Society.

Core Assets of ICAR

It has a vast network of 96 ICAR institutes, 77 All India Coordinated Projects/Networks, four deemed to be universities, two Central Agricultural Universities and 641 Krishi Vigyan Kendras (KVKs) spread across the country. In addition, there are 62 state Agricultural/Veterinary/Horticultural/Fishery universities and 4 general universities with agricultural faculty, as part of the NARES. Extensive research infrastructure representing many agro-ecosystems and highly prized genetic resource apart, there is a critical mass of highly accomplished scientists with multidisciplinary knowledge. The Council has developed a fairly good national (SAUs, CSIR, IITs and industry) and international (CGIAR institutes, ACIAR, European Union etc) collaborative network. The public support and appreciation for the ICAR, by way of favourable research policies, funding and recognitions, is steadily growing.

Past Achievements

“The ICAR has served our country with great distinction ----- . The contribution of ICAR scientists in the achievement of national self-sufficiency in food grains and diversity in food production is truly enormous.”

Prime Minister of India, 16 July 2011

The past research contributions of ICAR and its partners have been exceptionally higher than of a similar organization in any other developing country in the world. The empirical evidence has shown that investment in agricultural research and development in India has been a win-win option as it was the largest contributor to the agricultural total factor productivity, which in turn significantly contributed to reducing rural and urban poverty (NAAS, 2009). During 1980-81 to 2006-07, the average internal rate of return to the investment in agricultural research was about 46 per cent, which is comparable to that obtained internationally (Chand and Kumar, 2011). The growth in total factor productivity (TFP) between 1980 and 2000 was estimated to be in the range of 1.4-2.0 per cent, which is equal to the growth observed in crop sector during the green revolution period. During post-green revolution period of 2000-2008, the TFP rose to 3.06 per cent which is commensurate with the growth in crop sector during the same period (Rada, 2013). The research programs under the umbrella of ICAR are designed and undertaken for harnessing the power of science and technology that promotes food, nutritional and livelihood security of vast population of our country. In the past, ICAR played an enabling role in ushering green revolution and also in the subsequent developments in agriculture. The ICAR along with its partners in the NARES through research and technology developments,

enabled the country to increase the production of foodgrains by 5-fold, horticultural crops by 6-fold, fish by 12-fold (marine 5-fold and inland 17-fold), milk by 8-fold, and eggs by 27-fold since 1950-51. This sterling progress in production and productivity has made a visible impact on the national food and nutritional security. It has also played a major role in promoting excellence in higher agricultural education. The Council is engaged in cutting-edge science and technology development, and is internationally acknowledged for its leadership in the developing world.

The Council acknowledges the importance of partnerships and synergies of different stakeholders in providing technological solutions for agriculture. It has developed the organization's Intellectual Property Rights (IPR) domain, and has implemented its guidelines for Intellectual Property Management and Technology Transfer/Commercialization (ICAR, 2006). It is expanding its reach for generating and disseminating new knowledge across its wide range of stakeholders in the production and food value-chain. It is strengthening its partnerships with national and international organizations, various government departments, farmers and farmers' organizations, non-governmental organizations and the private sector involved in agri-business. Recognizing the importance of commercialization of innovations in partnership with industry, the ICAR has established 'Agri-Innovate India Ltd', a corporate entity for technology commercialization and consultancy services at home and abroad.

ICAR Today

- Leads with distinction one of the largest national agricultural research systems.
- Custodian of a huge collection of germplasms of plants, animals, fishes and microbes for future technology generation.
- Provides science-based agro-technology advisory services to the large farming community, free of charge.
- Supports agricultural education in the country through its deemed to be University Institutes and State Agricultural/Veterinary/Horticultural/Fishery universities on the pattern of University Grants Commission.
- Provides evidence based techno-economic recommendations to the Government of India for policy formulation for agriculture sector.
- Provides techno-economic and educational support to developing countries for increasing south-south cooperation.

The ICAR considers Indian farmers as its main client and benefactor, and is fully engaged with them. Towards this end the National Bioinformatics Grid and mobile extension services for a comprehensive approach from 'seed-to-market' have been put in place.

The ICAR/DARE is the first ISO-9000-2008 compliant Department as designated by the Quality Management System (QMS).

ICAR-2050

Globally, agricultural science, natural environment and the society are undergoing rapid transformations. ICAR is committed to reinvent, redesign and transform so as to address the complex changes and harness the domestic and global opportunities to become a leading agricultural research organization, which is responsive and sensitive to the needs of stakeholders.

Vision

Lead India towards attaining sustainable food, nutritional, environmental towards and livelihood security through agricultural research and education.

Mission

Harness the power of science and innovation for food security, food safety, farmer prosperity and enhance natural resources base to promote inclusive growth and sustainable development .

Guiding Principles

The ICAR accords highest priority to the changing needs of farmers of 2050; its entire philosophy is based on ‘farmer first’ which is part of its motto, with due consideration for other stakeholders in the entire value chain. The guiding principles that will provide direction to ICAR for determining its future research and education investment priorities and strategic framework are outlined in Box 2.

Focus Areas

The Indian National Agricultural Research and Education System (NARES) has to find solutions to the immediate problems of farming as well as keep its competence in technology development in the forefront, to address the continuously emerging problems. A National Fund for Basic and Strategic Research is functioning under the ICAR, with the objectives of building capacity and breaking yield and quality barriers through the partnerships of all. The Council is embarking on two missions, - *‘Farmer FIRST’* and *‘Student READY’*, as also the consortia platforms for bringing greater partnerships. Two major projects, viz. National Agricultural Education Project and National Agricultural Entrepreneurship Project, are being formulated. Value-addition to primary agricultural production in crop/horticulture/dairy/meat/fish, in order to make the agricultural sector competitive, will be a priority. Mechanisms

Guiding Principles for Future Research and Education

- Provide leadership in ensuring national food and nutritional security, farmers' prosperity, consumer health and enhancing the natural resource base of agriculture for future generations
- Ensure strategic competitive advantage of Indian agriculture to enable access to the existing and emerging markets, and address the emerging challenges
- Leverage the advances in other sciences, engineering and social science to enhance agricultural research
- Nurture scientific excellence and promote interdisciplinary, systems-based, knowledge-intensive, problem-solving research
- Promote economic opportunities for the rural community and society
- Promote complementary partnerships for value addition in agriculture and accelerate innovation
- Respond proactively to farmers, consumers, partners and policy makers
- Promote ethical conduct, scientific integrity and accountability of performance and decisions
- Promote organizational transformation to an efficient, effective, and responsive innovation system
- Support higher education and create educational environments that foster continuous learning

are also being developed for Agri-Innovation fund, Agri-Incubation fund, Regional implements fabrication and processing hubs, and Referral facilities. The major focus areas of research are outlined in Box 3.

ICAR 2050: Focus Areas of Research

- Genetic potential enhancement of agricultural commodities
- Agricultural productivity, efficiency and profitability improvement
- Resilience to climate change and abiotic and biotic stresses
- Improve nutritional food, and health security
- Risk management against climate change and market stressors
- Agricultural value chains
- Sustainability of natural resources base of agriculture
- Valuation of ecosystem services
- Agricultural markets, policies, and institutions
- Bio-security, especially the one emerging from gene piracy and cross-border vector-borne diseases
- New products and uses (eg, bio-energy, new crops, synthetic foods, special foods)
- New educational and learning systems and environments

Some transformation in intra-institutional arrangements for more efficient governance within the ICAR and the ICAR vis-a-vis state agricultural universities, as per need of the system will be required. The ICAR system has shown resilience in the past and has mechanisms in place for course correction.



Harnessing Science and Innovations

The ICAR is in the process of charting an innovative research agenda to fulfil its mission within the broad emerging science and technological trajectory, which is in harmony with nature and the goals and aspirations of society. Advances in science and technology help solve problems that have been in the realms of imagination. The Council has a vast mandate dealing with agriculture and food spanning across crops, natural resources, livestock and fisheries, and with all links in the value chain from primary production to processing – packaging – storage, etc. The technologies —new, currently in-use, in pipeline and awaiting harnessing — fall in three categories: generic or platform technologies (bio-, nano, info, and geospatial technologies), sector-specific technologies, and trans boundary research. There is a growing need to move towards trans disciplinary research, which has more to do with transgressing boundaries among disciplines and actors leading to a ‘fusion’ among disciplines and partnerships. The future research needs to support a profitable, more productive, and sustainable agriculture; laying emphasis on food and farm technologies that mitigate climate change impact and adapt to a low-carbon and low-water regimes, and highly modified biological responses to climate changes. The research vision in key areas of technology development are briefly introduced.

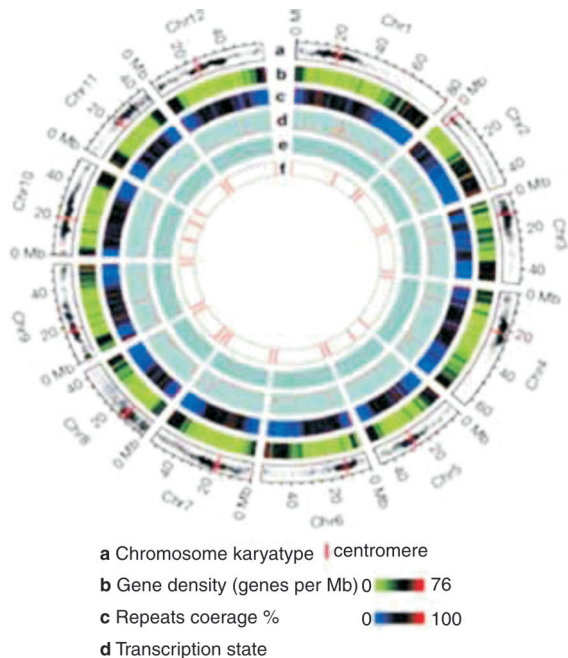
(1) Genetic enhancement of plants/animals/fish for higher productivity under increased intensity of biotic and abiotic stresses

Genetic enhancement is considered to be a major option to bridge the demand and supply gap under normal situations as well under projected scenarios of increased frequency and intensity of stresses. In agriculture, biotechnology has enabled the genetic alteration of crops, improved soil productivity, and enhanced natural weed and pest control. It should, however, be understood that genetically modified organisms (GMOs) may help solve some very difficult problems such as salinity or drought, but they do not present a miracle solution; and they are not automatically synonymous with sustainable foods. Emphasis is needed on green biotechnologies (concerning plants and their growing) as well as on the white biotechnologies (primarily focusing on use of biological organisms to produce or manipulate things).

Some of the important research issues in plant science include: improvement in productivity and resilience to climate change impacts and

up scaling of technologies with the aid of bio-, nano, omic (genomics, proteomics and metabolomics etc) and geospatial approaches; deep sequencing-based, cost-effective and rapid diagnostic tools for DNA fingerprinting and IPR protection, genetic modification (GM) detection and plant taxonomic identification systems; DNA bar-coding in species delineation hyper spectral remote sensing; structural and functional diversity present in the both coding and non-coding regions of the plant genome; digitization of mapping and distribution of pests including insects, fungi, bacteria, viruses, nematodes and weeds of quarantine significance; development of designer grains/fruits/vegetables possessing desirable qualities for processing and increased shelf-life.

In the livestock and fishery sectors, research can be utilized in the frontier areas like stem cells, pharmacokinetics and nutrigenomics, transgenic animals, proteome analysis, siRNA technology, bio-sensor applications, targeted nano-delivery of drugs; IVF-ETT, etc. can be gainfully utilized for strengthening system efficiency. Some important issues for investigation are: livestock genetic improvement using phenomics, genomics and bioinformatics tools; breeding transgenic animals capable of producing tailor-made milk/meat to cater to the specific needs; understanding of the basis of genetic resistance in domestic species of livestock with DNA markers for disease-resistant



Potato genome for future use (Source: ICAR Annual Report 2012-13)

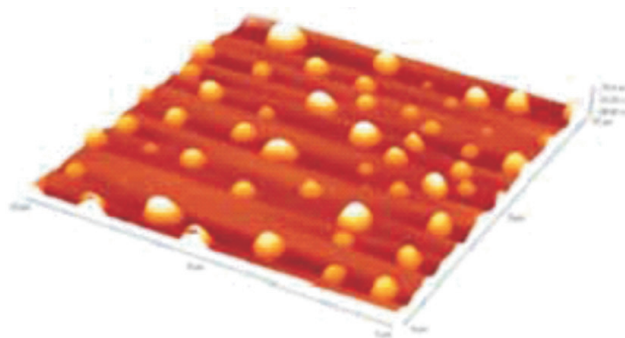
genes and the ability to diagnose specific genotypic markers that correlate with susceptible and resistant phenotypes; regenerative medicine, micro- electro- mechanical systems (Bio-MEMS), pharmacy-on-chip, implanted body regulator, gene-based preventive medicine and bio-electronics; efficient nutrient delivery in animals through application of nano-technology; and development and strengthening of bio-safety capabilities. Cytogenetics and genotoxicity of fish and shellfish; extraction and characterization of bio-molecules having therapeutic and industrial significance; identification of biosynthetic gene clusters in aquatic bacteria for production of novel bioactive compounds are some other frontier areas for research.

(2) Productivity enhancement through sustainable intensification

The earth's planetary resource system is reaching the tipping points in terms of anthropogenic eco-footprints; and under Business As Usual (BAU) scenario, it may require not one, but two such planets to remain within the safe operating space. Sustainable agricultural intensification, which holds promise in the twin roles — a major food security insurer and protector of natural resource base — has therefore rightly become central to the post-2015 research and development agenda.

Technologies and practices of sustainable intensification are context specific, and so are the research needs. It should, however, be understood that significant intensification is an option but only where enough water is available. Some broad issues are: a globally retrievable geo-spatial national portal on land, water, vegetation/animal/fish/microbes resources information system for resource inventorying and planning; understanding of enhanced agro-ecological adaptation of domesticated crop plants/animal/fish to abiotic and biotic stresses using novel and emerging technologies to develop low-input and low-carbon technologies; up scaling of water, nutrient, energy and knowledge smart technologies including conservation and precision agriculture to enhance land productivity in terms of kcal/ha/day; novel fertilizer formulations and delivery systems to improve nutrient-use efficiency, integration of crops, livestock, fish and forestry systems leading to higher productivity and lower negative impacts; waste management systems including extraction of nutrients from domestic and industrial waste waters, improved recycling of wastes to improve soil organic matter and soil health, microbial inoculants for biofertilization and disease control. The innovations in organic farming, particularly in dry lands and hills, where low input agriculture is practiced, are required for taking advantage of the growing market for organically grown food products.

Livestock feed, healthcare and management are to be addressed in an integrated manner for ensuring intensification in the animal sector. Changes in livestock production increase the potential for new pathogens to emerge, grow and spread from animals to humans on a global scale. Effective phytosanitary and animal health regimes aided by new vaccines, diagnostic products and tools, and epidemiological information are necessary to survive in the emerging regulatory regime.



Three dimensional image of 0.01per cent trypsin nano-encapsulated in **0.04 per cent chitosan** (Source: DARE/ICAR ANNUAL REPORT 2013–14,page 78)

The fisheries sector has shown a stellar growth in recent years, but recent studies cataloguing the global collapse of human-dominated marine ecosystems indicate that the marine fishery is experiencing accelerating loss of populations and species, with largely unknown consequences (Worm et al., 2006). Except for small



Cobia (Rachycentron canadum) has potential for cage farming

tank based aqua-culture, the situation is not different in case of riverine and coastal aqua-culture which suffer from high level of pollution. This is largely because aquaculture in most cases does not have dedicated water bodies. But, the good news is that fish stocks can recover, if ecosystems are protected and biodiversity gets revived. Therefore adoption of ecosystem-based approach to coastal as well as riverine aquaculture and diversification in terms of species, low in the food chain and developing aquaculture systems which are based on biofloc technology- utilizing the nitrogenous waste it produces, would be most appropriate.

(3) Productivity enhancement through mechanization of agriculture and food system

The end objective of farm mechanization is to enhance the overall productivity to save labour and automate field operations for lowering the cost of production through improved timeliness of operations and precision in the application of inputs; producing high quality and value-added products; and developing and saving energy. Intelligent sensing and monitoring systems equipped with crop and bioinformatics will be the new innovations in machines used for different operations. Machines for large-scale precision farming green fishing vessels’ will have to be developed. In the post-field state, logistics, large-scale processing, and supply chain management are the key factors. Information technology in association with GIS and automation are set to make a major difference at different stages (Fig 2). Optimization of production systems and farm work sites with vehicle and machine systems that can sense and control; systems that are capable of collecting, storing, and transferring information about the crop, field, and machine state at the time of field operation are likely to materialize in the country towards the end of the 4th decade.

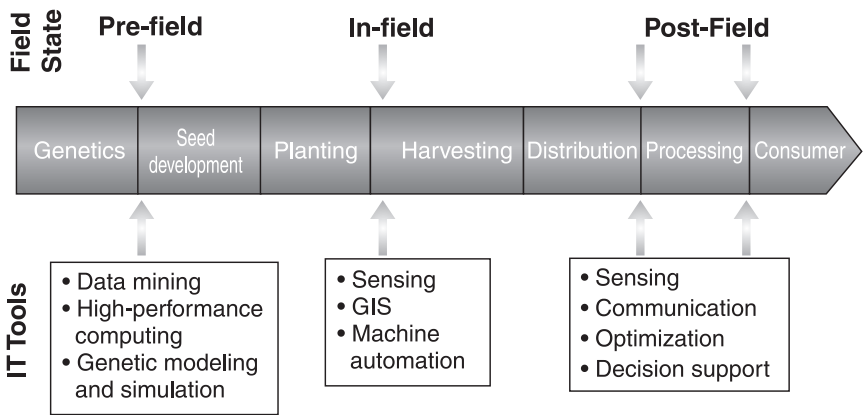


Fig. 2 Harnessing innovations in computing science, geospatial sciences and information technology (Source: Ting et al., 2011)

(4) Enhancing value, safety and income through food processing

Globally, India is a major producer of fruits, vegetables, milk, meat and fish, which are all perishable in nature. But the level of processing in India is rather low, and exists in unorganized form with high losses in supply chain. The emergence of new tools, methods, techniques and

approaches offers promise of technological breakthroughs to augment food security and income from agriculture enterprise by increasing secondary, tertiary and quaternary processing levels to at least 50 per cent. New opportunities are emerging with the opening of the international trade. Therefore, issues related to sanitary and phyto-sanitary measures would need to be appropriately addressed. Multi-disciplinary and multi-stakeholder research for agri-commodities, especially post-harvesting engineering, horticulture, dairy, livestock and fish, would be required.

Life cycle assessment (LCA) of products and processes in harvest and post-harvest constitutes an important area of future investigation. As value chains and markets get better organized and demand for specialized processed food grows, the focus will gradually move from primary and secondary processing to higher level of processing. Some important issues needing attention are: development of a library with documentation of desirable characters/attributes suitable for processing in developing designer crops; intelligent cool chain systems to transport produce from farm to retailer; adaptive control of storage conditions with biological sensors; rapid detection of food adulterants, fungal and bacterial toxins and other contaminants, using bio-sensors/nano-bio-sensors/molecular markers; application of robotics, artificial neural networking, nutri-genomics, non-destructive and/or online testing techniques, supercritical fluid extraction for production of high-value products; ethical and safety issues related to GM foods and nanotechnology; and development of bio-polymers for packaging and bio-composites for structures based on nano-technological development. The private sector has made considerable advances in food processing and therefore exploration of niche areas only would be more rewarding.

(5) Energy development and management

Energy is central to the economic growth in all the sectors including agriculture. In view of the dwindling and limited availability of fossil fuels, plants which exist through photosynthesis are a renewable source of carbon. Bio-fuels, therefore, appear to be part of the solution to the problem of energy shortage, but the development of more efficient bio-fuels remains a crucial issue. The agricultural fuel resources include animal manure and crop residues, derived primarily from maize, corn, small grains and seeds of oil-bearing plants (e.g. *Jatropha*). In India, bio-fuels would largely be based on nonfood crops and unused crop materials containing lignocellulose (stems and leaves). Therefore, development of engineering plants of different capacities, using waste material with lignocelluloses, is a priority. Researches in solar energy,

aiming at increased capture of solar radiations are opening vast opportunities for increasing availability of nonpolluting energy at low cost. In future, decentralized solar power for agricultural applications like pumping water, drying of grain and fruits, chilling of milk, will be financially attractive propositions. Efforts at innovations in development of appliances for rural areas need attention.

(6) Computational initiatives in agricultural research

Agricultural research is becoming highly quantitative and computational. Therefore, high performance computing (HPC) is becoming a requirement for manipulation of very large data sets, particularly related to agricultural genomics, proteomics, geo-informatics and climate change. An 'Integrated National Agricultural Resources Information System' (INARIS) and 'Knowledge Management of Agricultural Research and Technologies' (KMART) portals have been established. But, as the demand for national agricultural portal on bio informatics grid (NABG) grows, creation of higher super computing framework will have to be undertaken.

(7) Innovations in policies and institutions

Harnessing advances in agro-biological research in the form of technologies requires a favourable policy environment and the facilitating institutions. Research needs to reframe the conceptual maps regarding agriculture, environment, competitiveness, markets and state interventions. Further, agricultural research policy needs to address the mechanisms of generation of new sources of knowledge. Technology forecasting is of utmost importance. Foresight on research priorities, potential impacts of emerging technologies and consumer preferences in technology generation process, intellectual property regimes and seed systems, etc. needs to be generated. It would also be useful to develop agribiome platform (it takes into account the totality of food biomasses and uses calories as units), to generate information on regional food production, trade and uses of biomasses on eco-regional basis. All these issues require effective and need-based institutions to accelerate innovations and link farmers with different stakeholders to harness the emerging opportunities. Innovative institutional models, pro-agricultural policies and regulatory mechanisms would have to be evolved for accelerating innovations.

(8) Education and human-resource development

Higher education is crucial to maintain and sustain growth in

agricultural productivity, as it promotes research capabilities and skills to assimilate and adapt the current technologies, and develop the new ones. Enhancement of NARES' human resources for agricultural research and development is a priority issue for the Council. Considering the fact that globally the state of agriculture is in a flux, the ICAR institutes need to be geared up to respond to new and rapidly changing economic, ecological and technological environment. There are concerns in respect of both quality as well as the number of qualified human resources to meet the exploding demand. There is a need of paradigm shift in education — from 'teaching to learning', and of expanding the reach and opportunity for learning which will necessitate going beyond the current initiative of establishing new central agricultural universities and upgrading some of the existing institutes as deemed to be universities.

Agricultural research, education and extension for development (AREE4D) should be mainstreamed into national agricultural policy. There will also be a need to change the way in which we do science so that we may respond faster to societal demands. This may require re-structuring of the organisational model of research units, divisions from rigid ones to flexi-program mode dynamic research consortia led by program leaders on the pattern of international organisations. There is an urgent need to leverage developments in technology to improve the outreach and delivery of education. In future, improvement and expansion of platforms such as AgEd Open Course ware, International Platform for Asian Agricultural Education and E-Learning to meet the demand for formal and informal education for capacity building, will be undertaken.

(9) Innovations in technology transfer system

The ICAR has been engaged in development of efficient and effective technology transfer systems for shortening the time lag between technology generation and adoption. The motto of the Council, "*Agri Search with a Human Touch*", speaks volumes of the importance that the Council attaches to transfer of technology to its main client— the farming community. But, its clientele now extends to industry, business community and the consumers at large. There have been continuing efforts to evolve innovative pathways for disseminating breakthroughs in technology to farmers and the industry such as Agricultural Technology Management Agency (ATMA), Agri-clinics, Agri-business centres, Krishi Vigyan Kendras, E-chaupals, etc. The capacity building of community-based organizations and farmers' groups for acquiring knowledge and its transfer is being put in place for a faster dissemination. The rapidly

extending mobile telephone network is facilitating the shift from kiosk based information dissemination to mobile-based communication. The bundling of services approach such as insurance and credit agencies participating in agro technologies and weather advisory service, which has been successful abroad, has good prospects. Information and communication technology (ICT) with 3G and 4G technologies which are capable of delivering detailed text and pictures through the mobile phones, would revolutionize the technology dissemination process. A number of knowledge platforms, such as Smart Knowledge Agriculture Corridor (SKAC), Agropedia, Directorate of Knowledge Management in Agriculture(DKMA), Agricultural Websites and Agriculture Knowledge are some of the new models of technology transfer, which will be out scaled in future for a faster dissemination among masses.



Opportunities & Strengths

India, at present is an agriculture-based transforming country, where agriculture does not have a major share in the GDP growth, but livelihood and poverty remain overwhelmingly rural. The Vision 2050 visualises India as a developed country with a bulging middle class having reasonably high income and more than 50 per cent people living in urban areas. Therefore, the research strategies have to, some extent, be aligned with the future development goals. The research initiatives will aim at: zero net land degradation, 20 per cent increase in total food supply-chain efficiency– reducing losses and wastages from field to fork; 20 per cent increase in water and nutrient efficiency in agriculture–more nutrition and crop per drop; and enhancing food safety.

The Council's strategy for revitalising research agenda to provide technology, policy and institutional support will be based on leveraging knowledge and core strengths of ICAR for transforming and creating new institutions and hubs of scientific excellence and attract the best talents to agricultural research. The five important objectives are: create and accelerate sustainable increases in the productivity and production; conserve, enhance, and sustainably use natural resources and biodiversity; incentivise value addition; increase income and livelihood opportunities; evolve evidence based recommendations for policy and institutional changes which will stimulate agricultural growth and equity; and promote excellence in agricultural research, education and extension for sustained food and nutritional security (AREE4FNS).

Important Issues Linked to Approaches

Following are some of the linked issues:

- Upstream research for advancing frontiers of science for up gradation of technology taking advantage of the unprecedented advances in biological (biotechnology) and allied disciplines (nano, info and geospatial).
- Product differentiation (identification, development and optimization of niche markets) for global competitiveness
- Strategic management and prioritization of research needs at all levels in the agricultural sector to ensure that research resources (human, financial and infrastructure) are allocated as per new priorities (e.g., stronger demand for competitive and quality-conscious agriculture).

- Strengthening collaboration and partnerships with CGIAR institutes, in-country science and technology institutions and private sector to capitalize on opportunities for collective action and pooling of innovative capacity in agricultural production and value chains.
- Commercialization of agriculture through value chains and well-functioning national markets, for increasing rural income and supporting large urban and industrial population.
- Reforming NARES by establishing centres of agricultural research excellence to address strategic priorities and effective networking of cutting-edge institutions for attracting and nurturing talent, restructuring of the administrative structure and decentralization and financial devolution; and overhauling of technical services on the lines of ARS.

Approaches and performance indicators

The strategy to achieve the mission AREE4FSN is based on earnestly engaging in the task of harnessing the power of new science and technology, through planning, designing, implementation, monitoring and evaluation of the research outputs. Their efficacy will be ultimately judged by the outcomes and impacts. The approaches and indicators of performance linked to the AREE4FSN goal/objectives are outlined in Table 3.

Table 3 Strategic framework for Agricultural Research, Education and Extension for Food and Nutritional Security (AREE4FSN)

Goal / Objective	Approach	Performance Indicators
Improve food, nutritional, health security	Sustainable intensification of production system to increase food production Developing agriculture for improved nutrition and health outcome — nutrients fortified crops, animal products and fish products Development of rapid disease diagnostic systems, and drug delivery techniques Improved processing, packaging, storage and transportation for reduction in food losses from farm to fork Better assessment of supply and demand drivers	Increase in per capita food and nutrients availability/AGDP Reduction in malnutrition/decrease in poverty over the base line(%) Reduced mortality of animal/fish (%).
Create and accelerate sustainable increases in agricultural productivity and production	Genetic improvement of the leading food crops/ livestock/fish resources using frontier scientific advancements in bio-, nano-, omic sciences. Introduction of crop and landscape smart machines for precision agriculture Development and out scaled adoption of water/ nutrient/energy/weather/knowledge technologies Improvements in livestock/fish husbandry Improvements in livestock/poultry health through efficient drug delivery system	Increased productivity by (X) per cent/ha/year Increased production by (Y) Mt/year Reduction in water, energy, carbon intensity per unit of production Increased production kcal/ha/year Eradication of some of the economically important diseases

Improve and sustain health of natural resources	In-situ and ex-situ conservation of biodiversity of plant/animal/fish genetic resources Innovations in water- and nutrient-use technologies Better understanding of the impact of climate change and devising strategies for its adaptation and mitigation Improvements in policies for accounting ecosystem services; ecological economics; trade in carbon credits; virtual water trade	Reduction in extinction rate number/million() Decrease in rate of water diversions from ground and surface sources Reduction in water or energy or carbon intensity per tonne of production Carbon sequestration (Mt CO₂e/ha)
Increase income and livelihood opportunities for rural population	Diversification of production systems integrating crop/livestock/forest/fishery Development of models for commercialization of smart value chains and markets; and organize intellectual property rights and benefit-sharing system Introduction of bio-safety and sanitation regimes through automation, and development of standards	Increase in farm income (Rs/ha/year) Increase in volume of commodities processed over base line (%). Density of markets/cold chain established,(No./thousand ha) Standards developed and implemented.
Promote policy and institutional changes that will stimulate agricultural growth and equity	Policies for commercialization of technologies and food processing Addressing market risks through improved market intelligence and insurance products Policies and institutional arrangements to address climate change impact; disaster relief and bio-security	Improved market and will get piloted Spread of voluntary purchase of insurance cover by the farming community
Promote excellence in agricultural research, education and extension	Establishing centres of excellence in education and academic research Undertaking organizational reforms Research priority setting for optimization of resources (human/infrastructure/financial) Institutionalization of management information system (MIS) Fostering collaborations and partnerships Introduction of performance-based work-culture Follow participatory approaches; use of ICT Develop effective delivery systems and models for engagement with stakeholders Increasing investment in AREE4D	Increase in research publications with high impact factor/patents/commercialized technologies Increase in technology adoption index Systems developed and implemented Increase in resource allocation(% AGDP)

The ICAR is basically a research organization; and therefore for retaining a sharp focus and research leadership, it is important that a proper balance is maintained between research portfolio and technology diffusion, which sometimes gets mixed with field extension. Further, for implementation of transformative research agenda more efficiently, some changes in intra-organizational set-up, inter-organizational linkages

and responsibilities, investment in research and funding support from public and private sources will have to be undertaken. The ICAR has institutionalized mechanisms for ushering in such changes and will be in a position to reform the system as per the changing needs in the country.



Epilogue

The Indian Council of Agricultural Research (ICAR) is committed to enhancing agricultural production and productivity in the country through science-led and technology-driven approach. The Council firmly believes that agricultural research is meaningful only if it addresses the challenges faced by the society. The outputs of research must translate into increased productivity, improvement in quality of life and creation of wealth for the society at large, and small farmers in particular. The unfolding data revolution allows us to assess our present state, fixes the 2050 goal posts and indicates the direction for movement. The science-based, action-oriented agenda outlined in this ICAR Vision 2050 document, which integrates all the four interdependent dimensions of sustainable development (economic, social, environmental and governance), will be vigorously pursued.

A core program in the efforts to secure national food security is the promotion of gene revolution aiming at lowering the net production costs, raising the yields and net farm incomes, reducing the use of pesticides and herbicides, and thereby lowering the consumer prices. The second component of national research program deals with the development of new replicable models of sustainable production operating within the safe operating space, and respecting planetary boundaries of earth resources system. It requires creation of a wider ecosystem which connects small farm holders with knowledge, technology, well-managed value addition and marketing network for operating sustainable and remunerative agriculture and food systems. Therefore, research efforts aimed at developing appropriate models for commercialization of agriculture through value chains and market linkages form the third arm of AREE4FNS.

It is being realized that generation of technological solutions to address the challenges in production and processing activities throughout the value chain and the cost at which these are created, depends on the capacity built in the system. Capacity building in the NARES therefore, is a significant focus area. The capacity of the ICAR in serving the society also depends on its financial resources, which are provided mainly by the government. There is a vast difference in the public spending on agricultural research in India and the developed countries like Americas, Australia and in the European Union. It is envisaged that

with development of partnerships in certain areas, the public funding support will get augmented by the investments to be made by the industry and agri-business. The research outputs of ICAR institutes will be benchmarked to assess the system's performance in the context of similar national and international organizations.

Innovations in every endeavor that the Council pursues will be the hallmark criteria, with a firm conviction that innovations enable the organizations to pursue their specified goals efficiently. Appreciating the fact the institutions provide the environment within which innovations spurt, appropriate policy and institutional innovations (issues such as strategic planning, priority setting, R&D management, financing, monitoring and evaluation, etc.), will also be accorded due attention.



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APPENDIX I

Major supply side drivers (social, economic, environmental and technological), for agriculture and food system research 2010-11 and 2050

Driver	2010/11	2050	Implications
Climate change	Impacts in terms of heat waves, erratic rainfall and extreme events are visible	The rise in atmospheric temperature beyond 2°C under Business as Usual (BAU) scenario	Reduced productivity, rise in sea level, acidification of ocean, increase in frequency of floods/droughts
Water resources	Water resources are under stress in terms of quantity as well as quality	The situation may go beyond control under BAU scenario	Need for water smart technology will grow
Land			
Cultivated land	142 Mha Close to 2 million ha agricultural land has been lost to urban & industrial sectors	150 Mha In the absence of land use planning, the situation will get further aggravated Under BAU, situation will get worsened	Need for vertical intensification will become urgent; and conservation agriculture will get prominence Increased efforts for designing and implementation of watershed programs
Land and Soil health	Progressive increase in soil erosion and chemical degradation, loss of organic matter		
Biodiversity	Species disappearing at the rate of more than 10/ million/year	Rise in extinction rate under BAU	Concerted efforts are needed to improve the situation
Technology			
Genetics, bio-technology, nutri-genomics, and systems biology.	Strong advancements in genomics and proteomics. Until now only few new varieties have been produced	This group of technologies will become the dominant means of increasing productivity The range of applications in agriculture and food systems, moreso in animal science will grow	Lead to transformative improvements in productivity and reduction in cost Some social health concerns may arise
Nano-technology	There are promising prospects		Mechanization will grow Higher energy-use efficiency
Appropriate farm machinery	The trend towards mechanization has already set in	Fast development of ICT-supported robotics and automation	
Demographic trends in rural areas	Aging of rural populations as youths migrate to cities		Reduction in agricultural labour availability
Alternative food chains	Process is at the beginning stage	Will grow rapidly	There will be better links with markets and increase in farm income
Inputs and energy prices	High	May go higher	Force adoption of efficient technologies



APPENDIX II

Food demand drivers and response strategies

Drivers Strategy	Increased food demand	Climate change impacts	Declining and degrading natural resources	Declining bio-diversity	Energy demand	Technology requirement	Investments
Genetic enhancement	✓	✓	✓	✓	✓	✓	
Intensification	✓	✓	✓	✓	✓	✓	
Value addition & marketing	✓				✓	✓	
Mechanization	✓	✓	✓		✓	✓	
Energy development & management	✓	✓	✓			✓	
Innovations in policies and institutions	✓		✓				✓
Innovations in HRD	✓					✓	✓
Innovations in technology transfer system	✓		✓			✓	

Note: ✓ Indicates that the requirements imposed by a particular driver can be met through the identified strategic actions



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